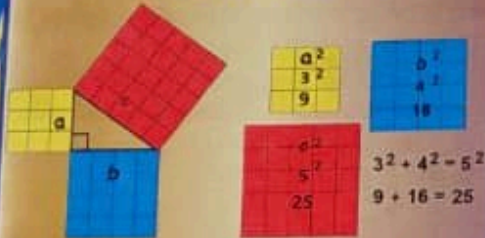


SHINE STANDARD MATHEMATICS

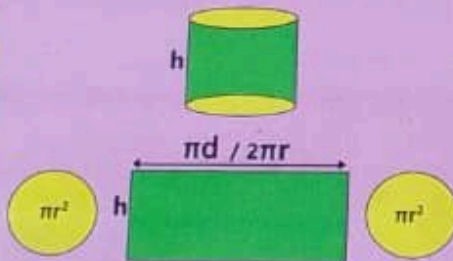
PUPIL'S BOOK 7

FIRST EDITION

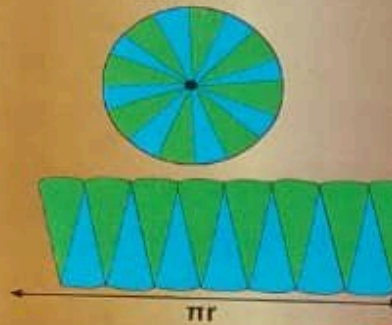
$$a^2 + b^2 = c^2$$



$$2\pi r^2 + 2\pi r h$$



$$\text{Area} = \pi r^2$$



For Excellence in Mathematics

ERT F.S.

BASED ON THE CURRENT CURRICULUM

SHINE STANDARD MATHEMATICS

PUPIL'S BOOK

FIRST EDITION

BY

AP SHINE STANDARD PUBLISHERS L

rights reserved. No part of this publication may be reproduced by any means, electronic, mechanical, photocopying, recording or otherwise, without the written permission of the publisher.

Copyright © AP Shine Standard Publisher



TABLE OF CONTENT
TERM ONE

Theme 1: Sets

- Topic 1: Set concepts**
- Finite and infinite sets
 - Subsets and proper subsets
 - Representing information on Venn diagrams
 - Solving word statements on Venn diagrams
 - Probability of sample events on Venn diagrams

Theme 2: Numeracy

- Topic 2: Whole numbers**
- Place values and values of whole numbers
 - Writing whole numbers in words and figures
 - Expanded form
 - Rounding off
 - Roman numerals
 - Bases

- Topic 3: Operation on whole numbers**
- The four basic operations on whole numbers
 - Properties of numbers
 - Indices/ exponents
 - Standard form/ scientific notation
 - Prime factorization of numbers

- Topic 4: Patterns and sequence**
- Divisibility tests
 - Types of numbers
 - Finding consecutive numbers

TERM TWO

- Topic 5: Fractions**
- Operations on vulgar fractions
 - Decimals
 - Ratios and proportions
 - Percentages
 - Simple interest

- Topic 6: Integers**
- Operations on integers
 - Finite/ modular system

Theme 3: Interpretation of graphs and data

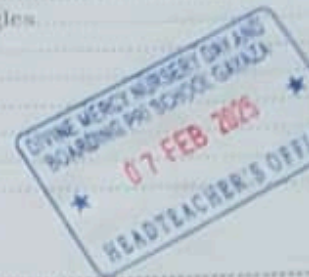
- Topic 7: Data handling**
- Measures of central tendency and range of grouped and ungrouped
 - Solving word statements on pie charts

- Distance time graphs
- Coordinate graphs
- Probability

Theme 4: Geometry

Topic 8: Construction

- Lines and line segment
- Complementary and supplementary angles
- Bisecting lines and angles
- Construction of special angles
- Construction of quadrilaterals
- Construction of regular polygons
- Triangulation
- Bearing and scale drawing



Theme 5: Measurement

Topic 9: Time

- Time in 12 and 24 hour clock system
- Distance, Time and Speed
- Average speed
- Time tables

TERM THREE

Topic 10: Length, mass and capacity

- Basic units of length
- Area and perimeter of figures
- Circumference and revolution
- Application of length
- Total surface area of solid figures
- Volume and Capacity of solid figures

Theme 6: Algebra

Topic 11: Algebra

- Collecting like terms and simplifying expressions
- Removing brackets and simplifying algebraic expressions
- Addition and subtraction of algebraic terms
- Solving equations
- Forming and solving equations
- Solving inequalities and finding solution sets

THEME 1: **SETS**

TOPIC 1: SET CONCEPTS

Remember:

- A set is a collection of well-defined members or elements.

Set symbols

Σ	Universal set
$\cap$	Intersection of sets
$\cup$	Union of sets
$n(P)$	Number of elements of set P
P^c	Complement of set P
$\subseteq$	Subset
$\in$	A member of a set
$\subset$	Proper subset

Ways of writing sets

- Sets are written in finite or infinite form

a) Finite sets

- Finite sets are sets with a definite or exact end.
- Finite sets are sets whose members can all be listed and counted.

Examples

- A set of odd numbers between 21 and 40
{23, 25, 27, 29, 31, 33, 35, 37, 39}
9 elements
- A set of alphabetical letters
{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z}
26 elements
- A set of whole numbers less than 10
{0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
10 elements
- A set of days of the week
{Sun, Mon, Tue, Wed, Thur, Fri, Sat}
7 elements

Note:

- ✓ All are finite sets because their elements have been listed and counted.

b) Infinite sets

- These are sets whose members cannot be all listed and counted.
- Infinite sets are sets with endless members or elements.
- Infinite sets are written with three (3) dots to show that the elements of that set are endless or continuing.

Examples

- ✓ A set of counting numbers
(1, 2, 3, 4, 5, 6, 7, 8, 9, ...)
- ✓ A set of integers
(..., 0, 1, 2, 3, 4, ...)
- ✓ A set of composite numbers
(4, 6, 8, 9, 10, 12, ...)

Note:

✓ All the above sets are infinite sets because their members cannot all be listed and the three dots show that the members are continuing.

Evaluation Activity



List the members of the following sets and state whether finite or infinite sets

- A set of even numbers between 15 and 38
- A set of odd numbers
- A set of natural numbers
- A set of multiples of 4
- A set of factors of 18

SUBSETS AND PROPER SUBSETS

- A subset is a set formed from the main/ given set
- A subset is a part of the main/ universal set
- A subset is a set contained in another set
- Subsets are categorized into two:
 - Proper subsets
 - Improper subsets
- A proper subset has less members/ elements than the given or universal set ($\subset$)
- An improper subset is a subset which is equal to the given or universal set ($\subseteq$)

Key points on forming subsets

- ✓ Every given set is a subset of itself.
- ✓ An empty set is a subset of every given set.
- ✓ Every given set has one improper subset which is the subset with the same elements as the given/ universal set.

Listing and finding number of subsets and proper subsets

Examples

- Given that set $P = \{a, g, t\}$, List and count all the subsets of set P.
(), {a}, {g}, {t}, {a, g}, {a, t}, {g, t}, {a, g, t}
Set P has 8 subsets
- Given that set $P = \{a, g, t\}$, List and count all the proper subsets of set P.
(), {a}, {g}, {t}, {a, g}, {a, t}, {g, t}
Set P has 7 proper subsets



3. Given that set $T = \{a, l, e, x\}$, List and count all the subsets of set T
 $\{\}, \{a\}, \{l\}, \{e\}, \{x\}, \{a, l\}, \{a, e\}, \{a, x\}, \{l, e\}, \{l, x\}, \{e, x\}, \{a, l, e\}, \{a, l, x\}, \{a, e, x\}, \{l, e, x\}, \{a, l, e, x\}$
 16 subsets
 Set T has 4 subsets
4. Given that set $T = \{3, k\}$, List and count all the proper subsets of set T
 $\{\}, \{3\}, \{k\}$
 Set T has 3 proper subsets.
5. The proper subsets of set R are $\{k, l\}, \{k, m\}, \{l, m\}, \{k\}, \{l\}, \{m\}$ and $\{\}$. Find $n(R)$.

Note:

- ✓ The original set is not regarded as a proper subset because proper subsets have less number of elements than the original

Finding number of subsets and number of proper subsets using the formula

Set	No of elements	Subsets	Power of 2	No of subsets
$\{\}$	0	$\{\}$	2^0	1
$\{l\}$	1	$\{\}, \{l\}$	2^1	2
$\{l, e\}$	2	$\{\}, \{l\}, \{e\}, \{l, e\}$	2^2	4
$\{l, e, x\}$	3	$\{\}, \{l\}, \{e\}, \{x\}, \{l, e\}, \{l, x\}, \{e, x\}, \{l, e, x\}$	2^3	8
	N		2^n	

- Number of subsets = 2^n , where n stands for the number of elements in a given set
- Number of proper subsets = $2^n - 1$, where n stands for the number of elements in a given set and 1 is the original set which does not qualify to be a proper subset

Examples

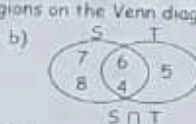
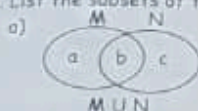
1. Given that set $P = \{a, g, t, y\}$, How many subsets are in set P ?
 Number of subsets = 2^n
 $= 2^4$
 $= (2 \times 2 \times 2 \times 2)$
 $= 16$ subsets
2. Given that set $P = \{a, g, t, y\}$, How many proper subsets are in set P ?
 Number of proper subsets = $(2^n) - 1$
 $= (2^4) - 1$
 $= (2 \times 2 \times 2 \times 2) - 1$
 $= 16 - 1$
 $= 15$ proper subsets
3. Given that set G has 5 elements, how many subsets are in set G ?
 Number of subsets = 2^n
 $= 2^5$
 $= 2 \times 2 \times 2 \times 2 \times 2$
 $= 32$ subsets

4. Given that set G has 5 elements, how many proper subsets are in set G?
Number of proper subsets = $(2^n) - 1$

$$\begin{aligned} &= (2^5) - 1 \\ &= (2 \times 2 \times 2 \times 2 \times 2) - 1 \\ &= 32 - 1 \\ &= 31 \text{ proper subsets} \end{aligned}$$

Evaluation Activity

- List all the subsets in the following sets
a) $B = \{1, 3, 4\}$ b) $G = \{r, k\}$ c) $H = \{b, e, d\}$ d) $F = \{\}$
- List the subsets of the given regions on the Venn diagrams below



- List all the proper subsets in the following sets
a) $F = \{d, 4\}$ b) $P = \{s\}$ c) $K = \{3, e, 6\}$
- Given the subsets of set W has $\{\}, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}$. List all the members of set W.
- The proper subsets of set V are $\{\}, \{d\}, \{a\}, \{n\}, \{d, a\}, \{a, n\}$. Write all the elements of set V.
- Using the formula 2^n , find the number of subsets in the following sets
a) $B = \{1, 3, 4, 6\}$ b) $G = \{\}$ c) $H = \{b, e, a, d, s\}$
- How many subsets can be got from a set with 6 elements?
- Using the formula $2^n - 1$, find the number of proper subsets in the following sets
a) $B = \{1, 3, 4, 6\}$ b) $G = \{\}$ c) $H = \{b, e, a, d, s\}$
- How many proper subsets can be got from a set with 6 elements?
- If set $K = \{a, b, c, d, e, f, g, h, i, j\}$ and set $L = \{a, e, i, o, u\}$, list the subsets of $K \cap L$.
- Find the number of proper subsets in set Y which is a set of all prime numbers between 30 and 45.

Finding number of elements when given number of subsets and proper subsets

Examples

- How many elements are in a set with 16 subsets?

Solution

$$2^n = \text{No. of subsets}$$

$$2^n = 16$$

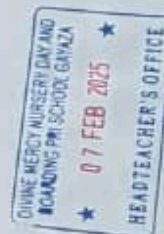
$$2^n = 2 \times 2 \times 2 \times 2$$

$$2^n = 2^4$$

$$n = 4$$

There are 4 elements

2	16
2	8
2	4
2	2
2	1



2. How many elements are in set with 31 proper subsets?

Solution

$$(2^n) - 1 = 31$$

$$2^n - 1 + 1 = 31 + 1$$

$$2^n = 32$$

$$2^n = 2^5$$

$$n = 5$$

There are 5 elements

2	32
2	16
2	8
2	4
2	2
	1

3. How many elements are in a set with { }, {1}, {2}?

Solution

$$(2^n) - 1 = n(\text{ })$$

$$(2^n) - 1 = 3$$

$$2^n - 1 + 1 = 3 + 1$$

$$2^n = 4$$

$$2^n = 2^2$$

$$n = 2$$

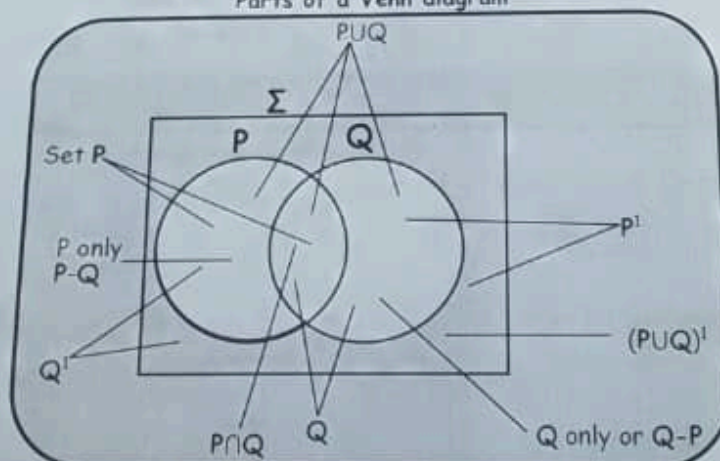


Evaluation Activity



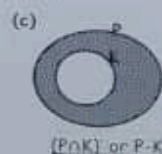
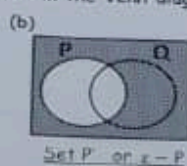
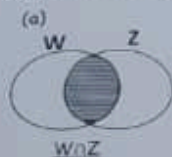
- Find the number of elements in a set with:
 - 4 subsets
 - 8 subsets
 - 16 subsets
 - 32 subsets
 - 64 subsets
 - 128 subsets
- Find the number of elements in a set with:
 - 3 proper subsets
 - 7 proper subsets
 - 15 proper subsets
 - 31 proper subsets
 - 63 proper subsets
 - 127 proper subsets
- Jamilah lists consecutive alphabetical letters starting from letter A. The set she forms has 32 subsets.
 - Find the number of elements in her set.
 - List all the elements in her set.

Parts of a Venn diagram



Examples

1. Describe the shaded part in the Venn diagrams below



2. Draw a Venn diagram to show that $A \cap B = A$



3. Draw a Venn diagram to show that $A \cup B = B$



4. Draw a Venn diagram to show that all dogs (D) are animals (A)



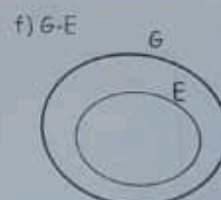
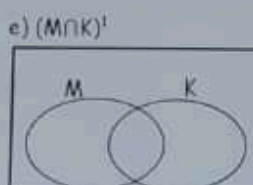
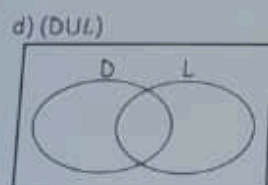
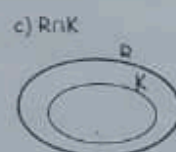
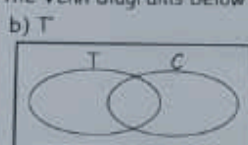
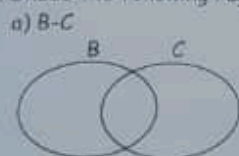
Note:

✓ This implies that dogs (D) are a subset of animals (A)

Evaluation Activity



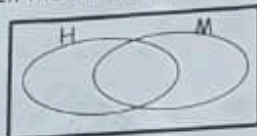
1. Shade the following regions in the Venn diagrams below



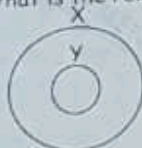
g) $(K \cup L)^c$



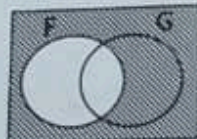
2. In the Venn diagram below, shade the complement of $H \cap M$



3. Draw a Venn diagram to show that all chickens (C) are birds (B)
4. Draw a Venn diagram to show that $R \cup H = H$
5. Draw a Venn diagram to show that $R \cap H = R$
6. What is the relationship between set X and Y below?



7. Describe the unshaded region.



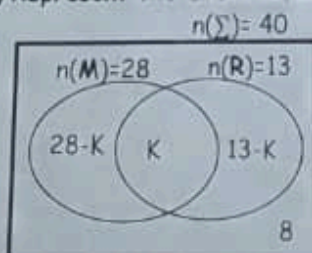
REPRESENTING INFORMATION AND SOLVING WORD STATEMENTS ON VENN DIAGRAMS

a) When the unknown is in the intersection

Examples

1. In a club of 40 farmers, 28 grow maize (M), 13 farmers rice (R), K farmers grow both maize and rice while 8 farmers do not grow any of the two crops.

a) Represent the above information on a Venn diagram.



b) Find the number of farmers who grow both crops.

Solution 1

$$28 - k + k + 13 - k + 8 = 40$$

$$28 + 13 + 8 - k = 40$$

$$49 - k = 40 - 49$$

$$49 - k = 40$$

$$-k = -9$$

$$\frac{-k}{-1} = \frac{-9}{-1}$$

$$k = 9$$

Solution 2

$$28 - k + k + 13 - k + 8 = 40$$

$$28 + 13 + 8 - k = 40$$

$$49 - k = 40$$

$$49 - 40 = k$$

$$9 = k$$

$$k = 9$$

c) How many farmers grow only one crop?

$$= (28 - k) + (13 - k)$$

$$= (28 - 9) + (13 - 9)$$

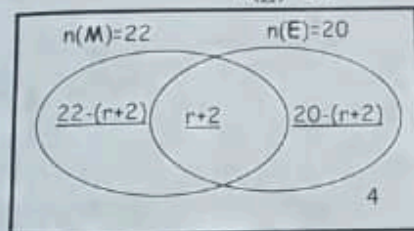
$$= 19 + 4$$

$$= \underline{23 \text{ farmers}}$$

2. In a class of 38 pupils, 22 like Math (M), 20 pupils like English (E), $r+2$ like both subjects while 4 pupils like neither of the two subjects.

a) Complete the Venn diagram below.

$$n(\Sigma) = 38$$



b) How many pupils like both subjects?

Solution

$$22 - (r+2) + r+2 + 20 - (r+2) + 4 = 38$$

$$22 - r - 2 + r + 2 + 20 - r - 2 + 4 = 38$$

$$22 + 20 + 4 + 2 - 2 - r + r - r = 38$$

$$46 - 2 - r = 38$$

$$44 - r = 38$$

$$44 - 44 - r = 38 - 44$$

$$\frac{-r}{-1} = \frac{-6}{-1}$$

$$r = 6 \text{ pupils}$$

$$r+2 = 22 + 20 + 4 - 38$$

$$r+2 = 46 - 38$$

$$r+2 = 8$$

$$r+2-2 = 8-2$$

$$r = 6$$

both subjects

$$r+2$$

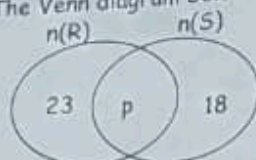
$$6+2$$

$$\underline{8 \text{ pupils}}$$

Evaluation Activity



1. The Venn diagram below shows the number of elements in set R and S.

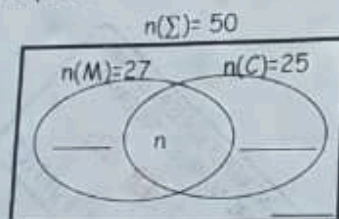


If $n(R \cup S) = 48$, find the value of p .

2. Given that $n(T) = 15$, $n(H) = 10$, $n(T \cap H) = 5$ and $n(T \cup H) = 4$

- Draw a Venn diagram showing the above information.
- Find $n(T \cup H)$
- Find $n(T \cap H)$
- Find $n(\epsilon)$

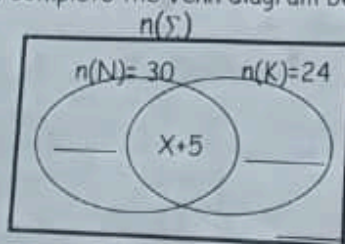
3. In a club of 50 farmers, 27 grow millet (M), 25 farmers grow cassava (C), 9 grow both millet and cassava while 9 farmers do not grow any of the two crops.



- Represent the above information on a Venn diagram.
- Find the value of n .
- Find the number of farmers who grow only one type of crop.

4. During a study tour, 30 learners visited Nile breweries (N), 24 learners visited a sugar factory (K), $x+5$ learners visited both, 14 learners visited Nile Breweries while 3 learners visited other places.

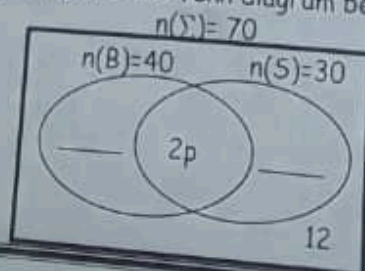
- Complete the Venn diagram below.



- Find the value of x .
- What is the probability that a learner at random visited both places?

5. At a party attended by 70 guests, 40 took beer (B), 30 took soda (S) 2p took both drinks and 12 guests did not take any of the two drinks.

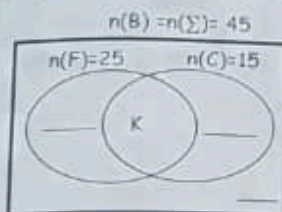
- Complete the Venn diagram below using the above information.



- Find the value of p .
- How many guests took only one type of drink?

6. In a class of 45 pupils, all of them eat beans (B), 25 eat fish (F), 15 eat Chicken (C), k pupils eat all the three types of food while 10 eat beans only.

a) Represent the above information on a Venn diagram



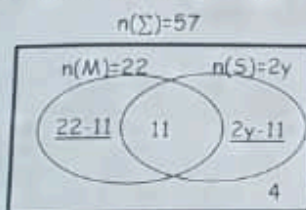
- b) Find the number of pupils who eat all the three types of food.
c) Find the probability of choosing a pupil who eats one type of food as a leader.

b) When one of the two sets is the unknown

Examples

1. In a team of 57 players, 22 players like music (M), 2y players like swimming (S), 11 players like both activities and 4 players do not like any of the two activities.

a) Complete the Venn diagram below



- b) Find the number of players who like swimming only.

Value of y

$$22 - 11 + 11 + 2y - 11 + 4 = 57$$

$$22 - 11 + 4 + 2y = 57$$

$$11 + 4 + 2y = 57$$

$$15 + 2y = 57$$

$$15 - 15 + 2y = 57 - 15$$

$$2y = 42$$

$$\frac{2y}{2} = \frac{42}{2}$$

$$y = 21$$

swimming only

$$2y - 11$$

$$(2 \times 21) - 11$$

$$42 - 11$$

$$31 \text{ players}$$

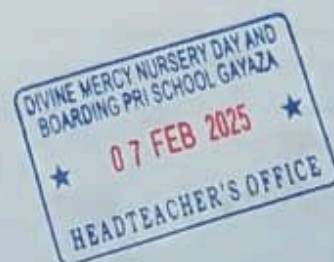
- b) Find the probability of picking a player who likes only one activity as a team captain.

$$\begin{array}{l} \text{One activity} \\ (22 - 11) + (42 - 11) \\ 11 + 31 \\ 42 \text{ players} \end{array}$$

$$\begin{array}{l} \text{probability} = \frac{\text{number of expected chances}}{\text{number of total chances}} \\ \frac{42}{57} \end{array}$$

Evaluation Activity

1. If $n(K) = 7$, $n(K \cap L) = 4$ and $n(L)_{\text{only}} = 8$, find:
a) $n(K \cup L)$
b) $n(L)$

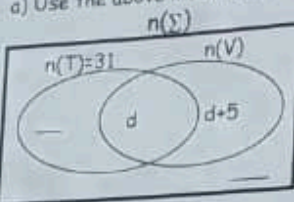


2. Given that $n(S)=19$, $n(T)=12$, $n(T \text{ only})=4$, $n(S \cap T)=7$ and $n(S \cup T)=r$.

2.

- Draw a Venn diagram and represent the above information correctly.
- Find $n(T)$.
- Find $n(S)$ only.

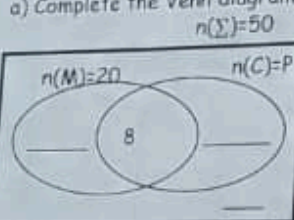
3. In a class, 31 pupils play tennis (T), $(d+5)$ play volley ball (V) only, d pupils play both games while 3 play neither of the two games.



3.

- If 27 pupils play volley ball, find the value of d .
- How many pupils do not play volley ball?

4. Of the 50 farmers, 20 grow millet (M), P farmers grow cassava (C), 8 farmers grow both crops and 6 farmers grow none of the two crops.



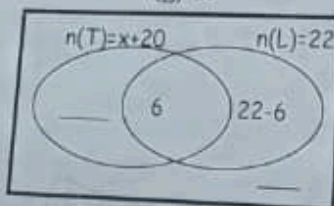
1.

2.

- How many farmers grow cassava only?
- Find the probability of choosing a farmer who grows millet only as a chairperson.

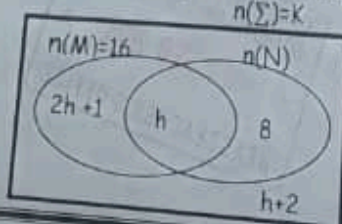
5. At a landing site, there are 53 fishermen, $x+20$ caught tilapia fish (T), 22 fishermen caught lung fish (L) and 6 fishermen caught both types of fish while 4 fishermen caught neither of the two types of fish.

a) Complete the Venn diagram below



- How many fishermen caught tilapia fish only?
- How many people caught lung fish but not tilapia?

6. Use the Venn diagram below to answer the questions that follow.



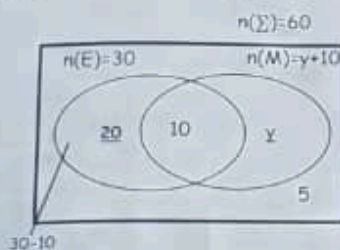
- Find the value of h .
- Find $n(N)$.
- Find the value of K .

c) When given complement or difference of sets as the unknown

Examples

1. In a class of 60 pupils, 30 like English (E) y like Mathematics (M) only, 10 like both subjects and 5 do not like any of the two subjects.

a) Use the information given above to complete the Venn diagram below



b) Find the value of y

$$y + 10 + 20 + 5 = 60$$

$$y + 35 = 60$$

$$y + 35 - 35 = 60 - 35$$

$$y = 25$$

c) How many pupils like only one subject?

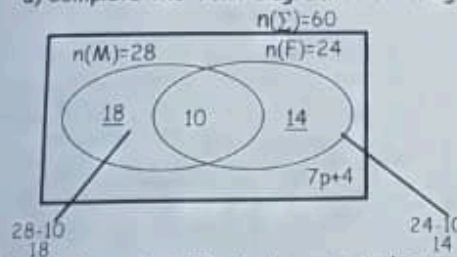
$$\text{Only one subject} = n(M) \text{ only} + n(E) \text{ only}$$

$$= 20 + 25$$

$$= 45 \text{ pupils}$$

2. On the teacher's day celebrations, 60 guests were invited and they were served with Fanta and Mirinda drinks. 28 were served with Mirinda (M), 24 were served with Fanta (F), 10 guests were served with both drinks while $7p+4$ guests were not served with any of the drinks.

a) Complete the Venn diagram below using the above information



b. Find the value of p.

$$18 + 10 + 14 + 7p + 4 = 60 \quad p = 2$$

$$46 + 7p = 60$$

$$46 - 46 + 7p = 60 - 46$$

$$7p = 14$$

$$\frac{7p}{7} = \frac{14}{7}$$

$$p = 2$$

c) Find the probability that a guest who was not served with any of the two drinks was the guest of honour.

$$\text{Not served any drink} = 7p + 4$$

$$= (7 \times 2) + 4$$

$$= 14 + 4$$

$$= 18 \text{ guests}$$

$$\text{Probability} = \frac{\text{number of expected chances}}{\text{number of total chances}}$$

$$= \frac{18}{60}$$

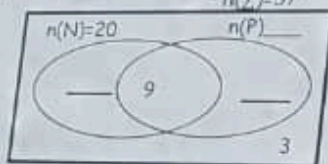
Evaluation Activity



1. In a class of the 57 attendants of the birthday party, 20 took Novida (N), $(2m+2)$ took Pepsi cola only (P), 9 took both types of drinks and 3 people did not take any the drink

a) Complete the Venn diagram below

$$n(S) = 57$$

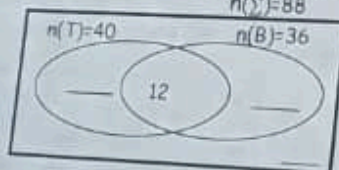


b. How many people took Pepsi cola?

2. In a group of 88 drivers, 40 drive taxis (T), 36 drive buses (B), 12 drive both taxis and buses while $4k+8$ do not drive lorries

a) Complete the Venn diagram below

$$n(S) = 88$$



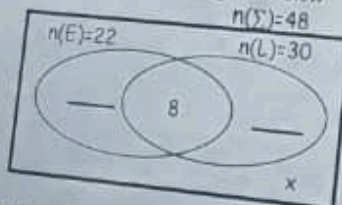
b) How many drivers do not drive taxis?

c) Find the probability of picking a driver who drives only one type of cars as a leader.

3. In a class of 48 pupils, 22 speak English (E), 30 speak Luganda (L), 8 pupils speak both the languages and x pupils speak neither of the two languages

b) Complete the Venn diagram below

$$n(S) = 48$$



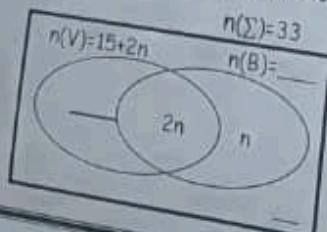
b) Find the value of x

c) Find the probability of picking a pupil who does not speak English from the class at random.

4. On a team of 33 players, 15 players play volley ball (V) only, n players play basketball (B) only, $2n$ play both games while 6 players play other games.

a) Use the above information to complete the Venn diagram below.

$$n(S) = 33$$

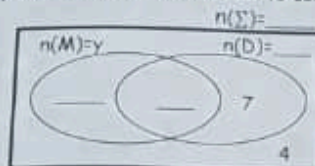


b) How many players play basketball?

c) What is the probability that a team captain plays basketball?

5. During an evening lesson, y pupils were taught by teacher Mary, 7 pupils were taught by only teacher Dissan, $(2y-18)$ pupils were taught by only teacher Mary and 4 pupils were not taught by any of the two teachers.

a) Use the above information to complete the Venn diagram below.

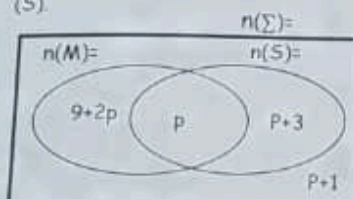


- b) If 5 pupils were taught by both teachers, how many pupils did teacher Mary teach?
c) How many pupils were taught altogether?

d) Solving word problems on Venn diagrams by comparing regions

Examples

1. The Venn diagram below shows learners who passed Mathematics (M) and Science (S).



- a) If 16 more learners passed Mathematics than Science, find the value of p .

$$n(M) - n(S) = 16$$

$$(9+2p+p) - (p+3) = 16$$

$$9+3p-p-3 = 16$$

$$9-3+3p-2p = 16$$

$$6+p = 16$$

$$6-6+p = 16-6$$

$$p = 10$$

- b) How many learners did not pass Mathematics?

$$n(M)' = (p+3) + (p+1)$$

$$= 10+3+10+1$$

$$= 24 \text{ learners}$$

- c) What is the probability that a learner who passed only one subject is a prefect?

$$\text{Only one subject} = (9+2p) + (p+3)$$

$$= (9+2 \times 10) + (10+3)$$

$$= 29+13$$

$$= 42 \text{ learners}$$

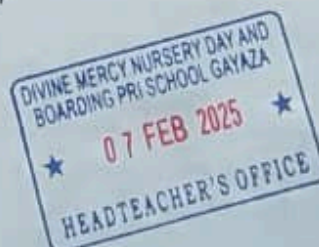
$$\text{Probability} = \frac{\text{number of expected chances}}{\text{Number of total chances}}$$

$$= \frac{42}{63}$$

$$n(S) = 9+2p+p+3+p+p+1$$

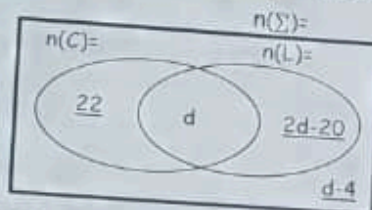
$$= 9+2 \times 10+10+3+10+10+1$$

$$= 63 \text{ learners}$$



2. In a bus 22 passengers bought chicken (C) only, d passengers bought both chicken and liver, $(2d-20)$ bought liver (L) only while $d-4$ bought neither of the two.

a) Complete the Venn diagram below.



- b) If the number of passengers who bought liver is twice the passengers who did not buy any of the types of food, find the value of d .

$$n(C \cup L) \times 2 = n(S)$$

$$2(d-4) = d + 2d - 20$$

$$2d - 8 = 3d - 20$$

$$2d - 3d - 8 = -20$$

$$-d - 8 = -20 + 8$$

$$\frac{-d}{-1} = \frac{-12}{-1}$$

$$d = 12$$

c) Find $n(S)$

$$22 + d + (2d - 20) + d - 4$$

$$(22 + 12) + (2 \times 12 - 20) + (12 - 4)$$

$$34 + 4 + 8$$

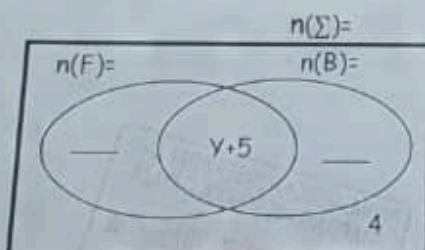
$$n(S) = 46$$

Evaluation Activity



1. In a certain class, 27 pupils play football only (F), $(y+10)$ play basketball only (B), $(y+5)$ play both football and basketball while 4 pupils play other games.

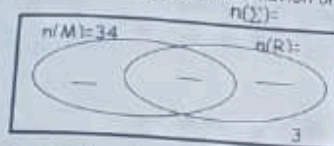
a) Complete the Venn diagram below.



- b) If 29 pupils play basketball, find the value of y .
c) Find the number of pupils in the class.

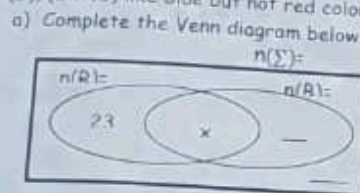
2. In a village, 34 farmers grow maize (M), k farmers grow rice (R) only, 8 farmers grow both and 3 farmers grow neither of the crops.

a) Represent the above information on the Venn diagram below.



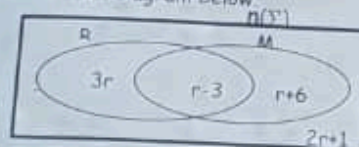
- b) Given that 23 farmers do not grow maize at all, find k .
c) How many farmers grow one type of crop?

3. In a group of workers, 23 workers like red colour only (R), x like both red and blue (B), $(2x+10)$ like blue but not red colour while three like none of the two colours.



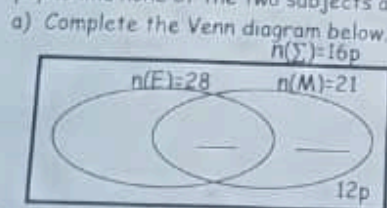
- b) If 43 workers like only one colour, find the value of x .
c) Find the probability of picking a worker who likes blue colour from the group.

4. At Mukozu village, some farmers grow beans (B) and others grow maize (M) as shown on the Venn diagram below.



- a) If 38 farmers grow only one crop, find r .
b) Find the total number of farmers who do not grow beans.
c) Find the total number of farmers in Mukozu village.

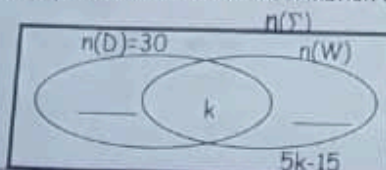
5. In a class of 16p pupils, 28 pupils like English (E), 21 pupils like Mathematics (M), 12p pupils like none of the two subjects and 3p like both English and Mathematics.



- b) How many pupils like both subjects?

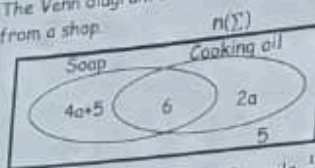
6. In a Primary Seven class, 30 pupils joined Debating club (D), 11 pupils joined Writers' club (W) only, k pupils joined both clubs while $(5k-15)$ pupils joined neither of the two clubs. If those who joined other clubs are twice those who joined both clubs,

a) Represent the above information on the Venn diagram below.



- b) Find the value of k .
c) Work out the probability of choosing a pupil who joined Writers' club.

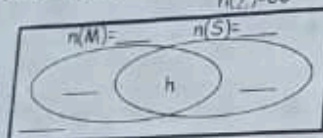
7. The Venn diagram below show the number of customers who bought different items from a shop.



- a) If 27 customers bought soap, find the value of a .
b) How many customers bought items from the shop?

8. A Primary Seven class has 60 pupils. $\frac{1}{3}$ of them like Mathematics (M), $\frac{1}{2}$ of them like Science, h pupils like both Mathematics and Science where as those who like neither the two subjects are equal to those who like Mathematics.

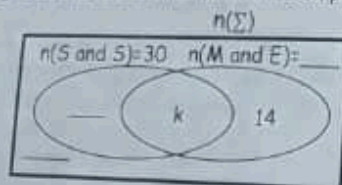
a) Use the above information to complete the Venn diagram.



- b) How many pupils like only Mathematics?

9. A class teacher issued out the end of year certificates to the outgoing candidates. 30 of them received a certificate each for being good at Science and Social Studies (S and S), 14 of them received a certificate each for being good at only Mathematics and English (M and E), k candidates got both certificates each while $3k+10$ pupils received certificates for being good at co-curricular activities.

- a) Use the above information to complete the Venn diagram.



- b) If $n(M \text{ and } E)=23$, find $n(c)$.

PROBABILITY

- Probability is the number used to describe the chance that something will happen or not happen.
- Probability is the likelihood of an event to happen

$$\text{Probability} = \frac{\text{number of expected chances}}{\text{number of total chances}}$$

$$= \frac{n(E.C)}{n(T.C)}$$

Examples

1. If a dice is tossed once, find the probability that a prime number will show on top.

$$\text{Probability} = \frac{n(E)}{n(T)}$$

$$T = \{1, 2, 3, 4, 5, 6\}$$

$$E = \{2, 3, 5\}$$

$$\text{Probability} = \frac{3}{6}$$

2. Alex is planning to go to Nairobi next week. What is the probability that he will go on a day that starts with letter "T"?

$$S = \{\text{Sun, Mon, Tue, Wed, Thurs, Fri, Sat}\}$$

$$E = \{\text{Tue, Thurs}\}$$

$$\text{Probability} = \frac{n(E)}{n(T)}$$

$$\text{Probability} = \frac{2}{7}$$

3. In a basket, there are 4 apples and 5 mangoes. Find the probability of picking a mango from the basket at random.

$$\text{No. of mangoes} = 5$$

$$\text{No. of Sample space} = (4+5) = 9$$

$$\text{probability} = \frac{n(E)}{n(T)}$$

$$\text{Probability} = \frac{5}{9}$$

10. The probability of Kareen visiting her uncle in the next holiday is $\frac{2}{3}$. What is the probability that Kareen will not visit the uncle?

Solution

$$\text{Probability} = \frac{1-2}{3}$$

$$\text{Probability} = \frac{3}{3} - \frac{2}{3}$$

$$\text{Probability} = \frac{3-2}{3}$$

$$\text{Probability} = \frac{1}{3}$$

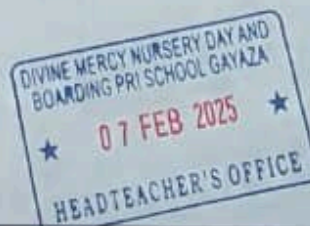
11. Toronto Community School will play a football match with Kampala Parents' School next week. What is the probability that Toronto Community School will win the match?

$$S = \{\text{win, loss, draw}\}$$

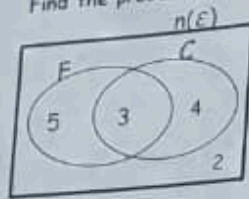
$$E = \{\text{win}\}$$

$$\text{Probability} = \frac{n(E)}{n(T)}$$

$$\text{Probability} = \frac{1}{3}$$



12. The Venn diagram below shows the number of pupils who in a class who eat fish (F) or chicken (C). Find the probability of picking a pupil at random who eats fish.



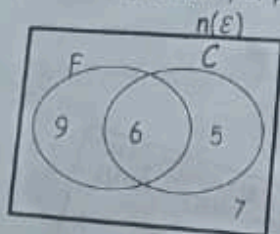
$$\begin{aligned}\text{No. of Events} &= 5+3 \\ &= 8\end{aligned}$$

$$\begin{aligned}\text{No. of Sample space} &= 5+3+4+2 \\ &= 14\end{aligned}$$

$$\begin{aligned}\text{Prob} &= \frac{n(E \cap F)}{n(E)} \\ &= \frac{8}{14}\end{aligned}$$

Evaluation Activity

- If a dice is tossed once, what is the probability that the following will show on top?
 - a prime number
 - a number less than 5
 - an even number
 - a composite number
 - a square number
 - a triangular number
- Juma is planning to travel to America next week. What is the probability that he will go on a day that starts with letter "S"?
- Gerald's wedding is next year. What is the probability that it will be held in a month with 30 days?
- In a basket, there are 4 oranges, 5 lemons and 7 apples. Find the probability of picking:
 - an orange at random?
 - a lemon at random?
- The probability that Moses will sit for an examination is $\frac{3}{4}$. What is the probability that he will not sit for an examination?
- Shine Junior School will play a football match with Kampala Parents' School next week. What is the probability that Excel Junior school will lose the match?
- The Venn diagram below shows the number of farmers who grow food crops (F) or cash crops (C).
 - Find the probability of picking a farmer at random who grows food crops.



- Find the probability of picking a farmer at random who grows cash crops.
- Find the probability of picking a farmer at random who grows only one crop.

THEME 2: **NUMERACY**

TOPIC 2: WHOLE NUMBERS

Place values of whole numbers

- A place value is the position of a digit in a numeral
- Place values help to identify positions of digits in given numerals.

Place values of whole numbers

Millions	Thousands	Units
Millions	Thousands	Ones
Ten millions	Ten thousands	Tens
Hundred millions	Hundred thousands	Hundreds

Examples

1. What is the place value of 9 in 249,367?

Solution

HTh	TTh	Th	H	T	O
2	4	9	3	6	7

The place value of 9 is Thousands.

2. Write the place value of 2 in 82,439,507.

Solution

TM	M	HTh	TTh	Th	H	T	O
8	2	4	3	9	5	0	7

82,439,507



Millions

The place value of 2 is millions.

3. The value of 6 in a number is 6000. What is its place value?

Solution

$$\begin{aligned}\text{Place value} &= \frac{\text{value}}{\text{digit}} \\ &= \frac{6000}{6} \\ &= 1000 \\ &\text{Thousands}\end{aligned}$$

Values of whole numbers

- A value is the quantity represented by a digit in a given number or numeral.
- A value is the product of the digit and its place value.

$$\text{Value} = \text{digit} \times \text{place value}$$



Place value in words	Place value in figures
Ones	1
Tens	10
Hundreds	100
Thousands	1,000
Ten thousands	10,000
Hundred thousands	100,000
Millions	1,000,000
Ten millions	10,000,000
Hundred millions	100,000,000

Finding values of digits in a given numeral

Examples

1. Find the value of 7 in 4276213.
 4,276,213
 ↓
 Ten thousands
 Value $7 \times 10,000$
 70,000

2. The place value of 5 in a number is millions. Find its value.
 $5 \times 1,000,000$
 5,000,000

Operations on place values and values

Example

1. Find the difference between the place value of 2 and the value of 8 in 32,486

Solution

Value of 2
 32486
 ↓
 Thousands
 1000

Value of 8
 32,486
 ↓
 tens
 8×10
 = 80

difference

$$\begin{array}{r} 1000 \\ - 80 \\ \hline 920 \end{array}$$

Evaluation Activity



- Find the place value of each of the digits with a ring in the following numbers.
 a) 47(6)5 b) 569(3)8 c) (4)0657
- Find the value of each of the digits with a ring in the following numbers.
 a) 37(4) b) 42(5)89 c) 4(6)9594
- Find the sum of the values of 4 and 6 in the number 63478.
- Workout the product of the values of 3 and 5 in the number 3765.
- Find the product of the value of 4 and the place value of 6 in the number 4516.

6. Represent the following numbers on an abacus.

- 4302
- 3205
- 2520

7. Given the place value of 8 in a numeral as Ten thousands, find its value.

8. The value of 3 in a numeral is 300, find the of its place value in the same numeral.

9. Find the quotient of the value of 6 and the place value of 9 in 46,293.

10. Given 947,856

(i) Find the product of the place value of 4 and the value of 6 in the above numeral.

(ii) Workout the sum of the value of 9 and the value of 8 in the above numeral.

(iii) Calculate the difference between the place values of 7 and 5.

11. The value of 3 in a number is 300 and the value of 2 in the same number is 2p. If the quotient of the value of 3 and 2 in the number is 15, find p.

12. The quotient of the value of 5 and the value of 2 in a number is 250. Given that the place value of 5 is thousands, find the value of 2.

13. Complete the table below given the numeral 89,573.

Digit	5		8	9	
Place value		Tens		Thousands	Ones
Value		70	80,000		3

14. The teacher wrote a number on the board as 6□□2. The tens and hundreds digits are missing. If $\frac{2}{9}$ of the calculated value the hundreds digit gives 150, what is the digit in the hundreds place value.

Writing whole numbers in words

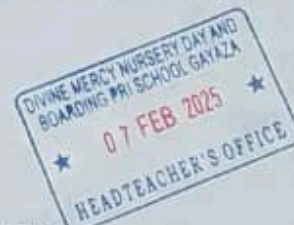
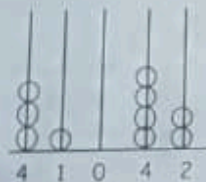
Examples

1. Write 92,407,013 in words.

Millions	Thousands	Units
92	407	013

Ninety-two million, four hundred seven thousand, thirteen

2. Write the number on the abacus in words



Thousands	Units
4 1	0 4 2

Forty-one thousand, forty two

Writing whole numbers in figures

- Group the number names according to the place values.
- Express into figures and arrange vertically.
- Add correctly to get the number in figures.

Examples

Write four hundred sixty-eight thousand, five hundred fifty-five in figures.

Solution

Four hundred sixty-eight thousand
Five hundred fifty-five

$$\begin{array}{r} 468000 \\ + 555 \\ \hline 468555 \end{array}$$

Mixed Activity

Writing whole numbers in words and figures

- Write the following in words.
 - 99040
 - 55009
 - 76,000,012
- Write the following in figures.
 - Forty-eight thousand thirty-six.
 - Seven hundred two thousand twenty-one
 - Ninety-one million one hundred thirty-four thousand nineteen
- Using the digits 3, 4 and 2
 - Write down all the possible 3-digit numbers that can be formed using the above digits.
 - Find the sum of the smallest and the largest three digit numbers formed.
- Given the digits 5, 3 and 0, write down all the 3-digit numbers formed using the given digits.
- Given the digits 4, 3, 1, 9, 8 and 7, Write the largest number formed using the given digits in words.
- Given the digits 3, 0, 5 and 2. Form the smallest four-digit number using the given digits.
- A pupil used the digits 3, 4 and 6 to write 3-digit numerals. Find the probability that he wrote an even number.

Writing numbers in expanded form using place values**Examples**

- Expand 469,017 using place values

HTh	TTh	T	H	T	O
4	6	9	0	1	7

$$(4 \times 100000) + (6 \times 10000) + (9 \times 1000) + (0 \times 100) + (1 \times 10) + (7 \times 1)$$

- Expand 571.62 using place values

H	T	O		Tth	Hth
5	7	1	.	6	2

$$(5 \times 100) + (7 \times 10) + (1 \times 1) + (6 \times 0.1) + (2 \times 0.01)$$

Finding the expanded number

Steps:

- Get the value for each digit being multiplied by the place value.
- Arrange the values according to place values vertically and find their sum.

Examples

1. Write $(4 \times 10000) + (8 \times 1000) + (2 \times 100) + (7 \times 10) + (9 \times 1)$ as a single number.

Solution

$$(4 \times 10000) + (8 \times 1000) + (2 \times 100) + (7 \times 10) + (9 \times 1)$$

$$40000 + 8000 + 200 + 70 + 9$$

$$40000$$

$$8000$$

$$200$$

$$70$$

$$+ 9$$

$$48279$$

2. What number has been expanded to give $(6 \times 100) + (3 \times 10) + (9 \times 1) + (4 \times 0.01)$

Solution

$$(6 \times 100) + (3 \times 10) + (9 \times 1) + (4 \times 0.01)$$

$$600 + 30 + 9 + 0.04$$

$$600.00$$

$$30.00$$

$$9.00$$

$$+ 0.04$$

$$639.04$$



Evaluation Activity



1. Expand the following numbers using place values.

a) 345681

b) 583241

c) 789240

d) 234807

e) 57913

f) 182.47

2. Find the numbers which have been expanded to give the following:

a) $(8 \times 1000) + (3 \times 10) + (6 \times 1) + (2 \times 0.1)$

b) $(5 \times 100) + (9 \times 1) + (4 \times 0.1) + (7 \times 0.01)$

3. Write as a single number.

a) $(3 \times 10) + (6 \times 1) + (5 \times 0.1) + (8 \times 0.01)$

4. Write in short.

a) $(9 \times 10) + (2 \times \frac{1}{10}) + (5 \times \frac{1}{100}) + (7 \times \frac{1}{1000})$

b) 4 thousands + 8 hundreds + 2 ones

Writing numbers in expanded form using values

1. Expand 93618 using values.

TTh	Th	H	T	O
9	3	6	1	8

$$(9 \times 10000) + (3 \times 1000) + (6 \times 100) + (1 \times 10) + (8 \times 1)$$

$$90000 + 3000 + 600 + 10 + 8$$

2. Express 5247.183 in expanded form using values

TH	H	T	O	TTh	Hth	THth
5	2	4	7	1	8	3

$$(5 \times 1000) + (2 \times 100) + (4 \times 10) + (7 \times 1) + (1 \times 0.1) + (8 \times 0.01) + (3 \times 0.001)$$

$$5000 + 200 + 40 + 7 + 0.1 + 0.08 + 0.003$$

Finding the expanded number

Arrange the given values according to place values vertically and find their sum

Examples

1. Write in short
90,000 + 6000 + 400 + 2.

Solution

$$\begin{array}{r} 90,000 \\ 6,000 \\ 400 \\ + 2 \\ \hline 96,402 \end{array}$$

1. Find the number which has been expanded to give:

$$300 + 8 + 0.4 + 0.06$$

$$\begin{array}{r} 300.00 \\ 8.00 \\ 0.40 \\ + 0.06 \\ \hline 308.46 \end{array}$$

Evaluation Activity

1. Expand the following numbers using values.

a) 342196

b) 437296

c) 389214

d) 359128b

e) 436.38

f) 802.167

2. Find the expanded numbers below:

a) $80,000 + 4000 + 900 + 20 + 6$

b) $4000 + 800 + 7 + 0.09$

c) $200 + 70 + 0.1 + 0.03$

d) $60,000 + 1000 + 200 + 40 + 0.8 + 0.07$

e) $90,000 + 200 + 50 + 8$

Writing numbers in expanded form using exponents or powers of ten

Examples

1. Write 236,804 as an expanded number using exponents.

HTh	TTh	Th	H	T	O
2	3	6	8	0	4
10^5	10^4	10^3	10^2	10^1	10^0

$$(2 \times 10^5) + (3 \times 10^4) + (6 \times 10^3) + (8 \times 10^2) + (0 \times 10^1) + (4 \times 10^0)$$

2. Expand 74,831 using powers of ten.

T	O		Tth	Hth	Thth
7	4		8	3	1
10^1	10^0	•	10^{-1}	10^{-2}	10^{-3}

$$(7 \times 10^1) + (4 \times 10^0) + (8 \times 10^{-1}) + (3 \times 10^{-2}) + (1 \times 10^{-3})$$

Finding the expanded number

Steps:

- Expand the power of ten for each digit.
- Multiply each digit by its expanded power of ten
- Get the value of each digit.
- Arrange the values according to place values vertically and find the sum.

1. Write $(8 \times 10^4) + (3 \times 10^3) + (0 \times 10^2) + (6 \times 10^1) + (2 \times 10^0)$ in short.

Solution

$$(8 \times 10^4) + (3 \times 10^3) + (0 \times 10^2) + (6 \times 10^1) + (2 \times 10^0)$$

$$(8 \times 10 \times 10 \times 10 \times 10) + (3 \times 10 \times 10 \times 10) + (0 \times 10 \times 10) + (6 \times 10) + (2 \times 1)$$

$$80000 + 3000 + 0 + 60 + 2$$

$$80000$$

$$3000$$

$$0$$

$$60$$

$$+ 2$$

$$\underline{83062}$$

2. What number has been expanded to give $(3 \times 10^2) + (6 \times 10^1) + (7 \times 10^{-1}) + (2 \times 10^{-2})$

$$(3 \times 10^2) + (6 \times 10^1) + (7 \times 10^{-1}) + (2 \times 10^{-2})$$

$$(3 \times 10 \times 10) + (6 \times 10) + (7 \times \frac{1}{10}) + (2 \times \frac{1}{10 \times 10})$$

$$300 + 6 + \frac{7}{10} + \frac{2}{100}$$

$$300 + 6 + 0.7 + 0.02$$

$$300.00$$

$$6.00$$

$$0.70$$

$$\underline{0.02}$$

$$\underline{306.72}$$

Evaluation Activity



1. Expand the following using powers of ten.

a) 234865

b) 542391

c) 345 718

d) 24.0539

e) 129.36

f) 76.428



2. Expand 74,831 using powers of ten.

T	O		Tth	Hth	THth
7	4		8	3	1
10^1	10^0	*	10^{-1}	10^{-2}	10^{-3}

$$(7 \times 10^1) + (4 \times 10^0) + (8 \times 10^{-1}) + (3 \times 10^{-2}) + (1 \times 10^{-3})$$

Finding the expanded number

Steps:

- > Expand the power of ten for each digit.
- > Multiply each digit by its expanded power of ten.
- > Get the value of each digit.
- > Arrange the values according to place values vertically and find the sum.

1. Write $(8 \times 10^4) + (3 \times 10^3) + (0 \times 10^2) + (6 \times 10^1) + (2 \times 10^0)$ in short.

Solution

$$(8 \times 10^4) + (3 \times 10^3) + (0 \times 10^2) + (6 \times 10^1) + (2 \times 10^0)$$

$$(8 \times 10 \times 10 \times 10 \times 10) + (3 \times 10 \times 10 \times 10) + (0 \times 10 \times 10) + (6 \times 10) + (2 \times 1)$$

$$80000 + 3000 + 0 + 60 + 2$$

$$80000$$

$$3000$$

$$0$$

$$60$$

$$+ \quad 2$$

$$\underline{83062}$$

2. What number has been expanded to give $(3 \times 10^2) + (6 \times 10^1) + (7 \times 10^{-1}) + (2 \times 10^{-2})$?

$$(3 \times 10^2) + (6 \times 10^1) + (7 \times 10^{-1}) + (2 \times 10^{-2})$$

$$(3 \times 10 \times 10) + (6 \times 10) + (7 \times \frac{1}{10}) + (2 \times \frac{1}{10 \times 10})$$

$$300 + 6 + \frac{7}{10} + \frac{2}{100}$$

$$300 + 6 + 0.7 + 0.02$$

$$300.00$$

$$6.00$$

$$0.70$$

$$\underline{0.02}$$

$$\underline{306.72}$$

Evaluation Activity



1. Expand the following using powers of ten.

a) 234865

b) 542391

c) 345.718

d) 24.0539

e) 129.36

f) 76.428



2. Find the numbers which have been expanded below.

- a) $(4 \times 10^3) + (9 \times 10^1) + (1 \times 10^2) + (6 \times 10^{-1}) + (3 \times 10^{-2})$
 b) $(7 \times 10^5) + (2 \times 10^4) + (8 \times 10^{-2}) + (6 \times 10^{-5})$
 c) $(5 \times 10^4) + (1 \times 10^1) + (3 \times 10^0) + (9 \times 10^{-2})$
 d) $(2 \times 10^1) + (8 \times 10^0) + (4 \times 10^{-1}) + (7 \times 10^{-2}) + (3 \times 10^{-3})$
 e) $(9 \times 10^4) + (3 \times 10^1) + (8 \times 10^0)$

ROUNDING OFF WHOLE NUMBERS

- Rounding off is a numeration system of approximating or estimating a number to its nearest quantity.
- Rounding off helps to create a sense of estimation
- We round off to a specified place value or a specified number of digits
- When a digit on the right of the required place value is 5, 6, 7, 8 or 9, we round up.
- And when the digit on the right of the required place value is 0, 1, 2, 3 or 4, we round down.

Rounding off scale

Less than 5	More than 5
0 1 2 3 4	5 6 7 8 9
0	1
Rounding down	Rounding up

Examples

1. Round off 493612 to the nearest thousands.

Solution

493612

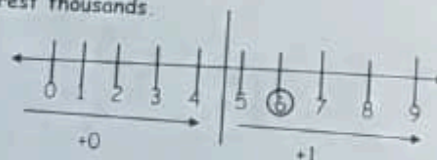
Required place value

493000

+ 1000

494000

493612 $\approx$ 494000



1 $\times$ thousands

1 $\times$ 1000 = 1000

2. Round off 6419 to the nearest hundreds.

Solution

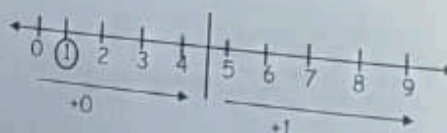
6419

Required place value

6400

+ 000

6400



0 $\times$ hundreds

0 $\times$ 100 = 0

Evaluation Activity



- Round off to the nearest tens.
a) 386 b) 4594 c) 999
- Round off to the nearest hundreds
a) 2721 b) 3582 c) 8457
- Round off to the nearest thousands.
a) 52941 b) 73061 c) 89603
- Round off to the nearest ten thousands
a) 95032 b) 21469 c) 16347
- A book costs sh. 27,500 at a shop. If the cost is rounded off to the nearest thousands;
a) will the buyers be happy or not?
b) Explain why.



ROMAN NUMERALS

- Roman numeral is a form of writing where numbers are expressed with special capital letters.
- They are only seven letters used in roman numerals and these form the basic roman numerals.
- Other categories of roman numerals include: repeated roman numerals, subtractional roman numerals and additional roman numerals

Basic Roman numerals

Roman numeral	I	V	X	L	C	D	M
Hindu Arabic number	1	5	10	50	100	500	1000

Repeated Roman numerals

Hindu Arabic number	2	3	20	30	200	300	2000	3000
Roman numeral	II	III	XX	XXX	CC	CCC	MM	MMM

NB: Each basic Roman numeral letter is repeated not more than thrice.

Subtractional Roman numerals

Subtraction	Hindu Arabic number	Roman numeral
5-1	4	IV
10-1	9	IX
50-10	40	XL
100-10	90	XC
500-100	400	CD
1000-100	900	CM

Additional Roman numerals

Addition	Hindu Arabic number	Roman numerals
5+1	6	VI
5+2	7	VII
5+3	8	VIII
50+10	60	LX
50+20	70	LXX
50+30	80	LXXX

Changing Hindu Arabic numerals to Roman numerals

1. Express 411 in Roman numerals.

Solution

$$411 = 400 + 10 + 1$$

$$CD \quad X \quad I$$

$$411 = CDXI$$

2. A farmer stored 1994 tonnes of rice in the store. Express the stored quantity in Roman numerals.

Solution

$$1994 = 1000 + 900 + 90 + 4$$

$$M \quad CM \quad XL \quad IV$$

$$MCMXLIV \text{ tonnes}$$

Evaluation Activity



- Write the following numbers in Roman Numerals
a) 54 b) 49 c) 99 d) 955 e) 415
- On a farm, there are 119 cows, 264 sheep and 186 goats. Express the total number of animals on the farm in Roman numerals.
- Alex sold 868 kg of maize. Write the quantity of maize he sold in Roman numerals.
- Given the number 64938,
a) write the sum of the values of 3 and 9 in Roman numerals.
b) Work out the difference of the values of 9 and 8 and give your answer in Roman numerals.
- The school enrollment was 754 last year and it increased to 1070 learners. Express the number of learners that increased in Roman numerals.

Changing Roman numerals to Hindu Arabic numerals

1. Write DCCCLXXXVIII in Hindu Arabic numerals.

Solution

$$DCCCLXXXVIII = DCCC \quad LXXX \quad VIII$$

$$800 \quad 80 \quad 8$$

$$888$$

2. Moses harvested MCMLVI litres of rain water and he sold DCCCLIX litres. How many litres of water did he remain with? Give your answer in Roman numerals.

Solution

$$MCMLVI = M \quad CM \quad L \quad VI$$

$$1000 + 900 + 50 + 6$$

$$1956 \text{ litres}$$

$$DCCCLIX = DCCC \quad L \quad IX$$

$$800 + 50 + 9$$

$$859 \text{ litres}$$

Difference

$$1956$$

$$- 859$$

$$1097 \text{ litres}$$

$$1000 + 90 + 7$$

$$MXCVII \text{ litres}$$

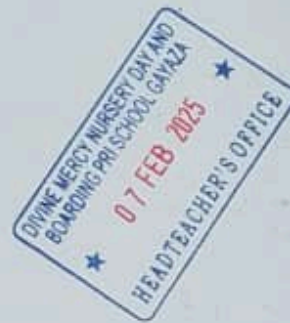
Evaluation Activity



- Write the following Roman Numerals into Hindu Arabic Numerals
 - LIX
 - XCIX
 - LXXXIV
 - MDCIV
- There were DCCXCIV books in the library. Write the number of books in Hindu Arabic numerals.
- Write MDCLXXXVII in Hindu Arabic numerals.
- Uganda got her independence MCMLXII. Express this year in Hindu Arabic numerals.
- Allan collected XCIX mangoes on Monday and Junior collected LXIV mangoes on the same day. Express their total collection in Roman numerals.
- Write DV in Hindu Arabic numerals.
- Complete the addition Roman table below.

+	IV	XXI	XIX
V		XXVI	
VIII	XII		

- Tell the morning time shown below.



BASES

Names of different bases

Base	Name	Digits used
Base two	Binary base	0,1
Base three	Ternary base	0,1,2
Base four	Quaternary base	0,1,2,3
Base five	Quinary base	0,1,2,3,4
Base six	Senary base	0,1,2,3,4,5
Base seven	Septenary base	0,1,2,3,4,5,6
Base eight	Octal base	0,1,2,3,4,5,6,7
Base nine	Nonary base	0,1,2,3,4,5,6,7,8
Base ten	Denary base/Decimal base	0,1,2,3,4,5,6,7,8,9

Place values in non decimal bases

Write the place values of each digit in the numbers below;

b) 101010_{two}

Solution

1 0 1 0 1 0 $_{\text{two}}$
 ones
 twos
 two twos
 two two twos
 two two two twos
 two two two two twos

a) 23102_{four}

Solution

2 3 1 0 2 $_{\text{four}}$
 ones
 fours
 four fours
 four four fours
 four four four fours

c) 76843_{nine}

Solution

7 6 8 4 3 $_{\text{two}}$
 ones
 nines
 nine nines
 nine nine nines
 nine nine nine nines

Values of digits in non decimal bases

Find the value of each of the underlined digits in the numbers below;

a) 110101_{two}

Solution

1 1 0 1 0 1 $_{\text{two}}$
 two twos $2 \times 2 = 4$
 two two two twos $2 \times 2 \times 2 = 16$

b) 45310_{six}

Solution

4 5 3 1 0 $_{\text{six}}$
 ones $1 \times 1 = 1$
 six sixes $5 \times 6 \times 6 = 180$

Evaluation Activity



- Write the place value of each of the underlined digits below:
 - 2011_{four}
 - 3022_{six}
 - 1111_{two}
 - 101101_{two}
 - 54362_{nine}
 - 24304_{five}
- Find the value of each of the underlined digits below:
 - 4263_{nine}
 - 4223_{five}
 - 2323_{three}
 - 10111_{two}
 - 11011_{two}
 - 6234_{seven}
- Show 403_{five} on the abacus.
- Represent the following on the abacus.
 - 231_{six}
 - 101_{two}

Converting/ changing base ten to non-decimal bases

Steps:

- Divide the given base ten number by the non-decimal base you are changing to.
- At every division record the remainder.
- After dividing, write the answer from the last remainder to the first and indicate the non-decimal base required.

Examples

1. Change 13_{10} into binary system.

B	N	R
2	13	1
2	6	0
2	3	1
2	1	1
	0	

1101

2. Express 152_{10} as an octal base numeral.

B	N	R
8	152	0
8	19	3
8	2	2
	0	

230_{eight}

Evaluation Activity

1. Convert 19_{ten} into base two.
2. Convert 19_{ten} into base two.
3. Express 33_{ten} to ternary base.
4. Group these discs in twos and write the number in binary system.
4. Convert 19_{ten} into base two.
5. Change 47_{ten} to quinary base
6. Write 162_{ten} as a nonary base numeral.



Changing/ converting non-decimal bases to decimal base (base ten)

Steps:

- Find the value of each digit in the given number.
- Get the sum of the values got and indicate base ten

Examples

1. Convert 213_{four} to base ten.

Solution 1

4^2	4^1	4^0
2	1	3

$$\begin{aligned} & (2 \times 4^2) + (1 \times 4^1) + (3 \times 4^0) \\ & (2 \times 4 \times 4) + (1 \times 4) + (3 \times 1) \\ & 32 + 4 + 3 \\ & 39_{ten} \end{aligned}$$

Solution 2

2 1 3 four
| | | ones
| | | four
| | | four

$$(2 \times 4 \times 4) + (1 \times 4) + (3 \times 1)$$
$$32 + 4 + 3$$
$$39_{\text{ten}}$$

2. Change 1111_{two} to denary base.

Solution 1

2^3	2^2	2^1	2^0
1	1	1	1

$$\begin{aligned} &(1 \times 2^3) + (1 \times 2^2) + (1 \times 2^1) + (1 \times 2^0) \\ &(1 \times 2 \times 2 \times 2) + (1 \times 2 \times 2) + (1 \times 2) + (1 \times 1) \\ &8 + 4 + 2 + 1 \\ &15_{\text{ten}} \end{aligned}$$

Solution 2

1 1 1 1 four
| | | ones
| | four
| fours
four four

$$(1 \times 2 \times 2 \times 2) + (1 \times 2 \times 2) + (1 \times 2) + (1 \times 1)$$
$$8 + 4 + 2 + 1$$
$$15_{\text{ten}}$$

Evaluation Activity



1. Convert the following into base ten (decimal base)
- | | | | |
|-------------------------|------------------------|------------------------|------------------------|
| a) 1011_{two} | b) 1101_{two} | c) 101_{two} | d) 111_{two} |
| e) 121_{three} | f) 202_{five} | g) 301_{four} | h) 123_{five} |

Changing non-decimal bases to non-decimal bases

Steps:

- Change the given non-decimal base to base ten.
- Change the answer got in base ten to the required non-decimal base.

Examples

1. Change 34_{five} to binary base

$$\begin{array}{r}
 34_{\text{five}} \\
 | \\
 \text{ones} \\
 \text{fives} \\
 (3 \times 5) + (4 \times 1) \\
 15 + 4 \\
 19_{\text{ten}}
 \end{array}$$

B	N	R
2	19	1
2	9	1
2	4	0
2	2	0
2	1	1
	0	

10011_{two}

Evaluation Activity



1. Change 103_{four} into binary base.
2. Convert 232_{four} to base five.
3. Convert 212_{four} into base five.
4. Change 243_{five} to octal base.
5. Convert 101_{two} into base four.
6. Express 64_{seven} as a nonary base numeral.
7. Change 121_{three} to base seven.

Finding the unknown base or number

1. Given that $21_n = 13_{\text{ten}}$, Find base n

Solution 1

$\begin{array}{ c c } \hline 2 & 1 \\ \hline n^1 & n^0 \\ \hline \end{array}$	$= 13$
$(2 \times n^1) + (1 \times n^0)$	$= 13$
$(2 \times n) + (1 \times 1)$	$= 13$
$2n + 1$	$= 13$
$2n + 1 - 1$	$= 13 - 1$
$\begin{array}{r} 12n \\ -1 \\ \hline \end{array}$	$\begin{array}{r} 12n \\ -12 \\ \hline 2n \\ -2 \\ \hline \end{array}$
n	$= 6$ n is base 6

Solution 2

$$\begin{array}{r}
 21_n = 13 \\
 | \\
 \text{ones} \\
 n \\
 2 \times n + 1 \times 1 = 13 \\
 2n + 1 = 13 \\
 2n + 1 - 1 = 13 - 1 \\
 \begin{array}{r} 12n \\ -1 \\ \hline \end{array} = \begin{array}{r} 12n \\ -12 \\ \hline 2n \\ -2 \\ \hline \end{array} \\
 n = 6
 \end{array}$$

2. Given that $32_y = 43_{\text{five}}$, Find base y

3	2	=	4	3
y^1	y^0		5^1	5^0

$$(3xy^1) + (2xy^0) = (4 \times 5^1) + (3 \times 5^0)$$

$$(3xy) + (2x1) = (4 \times 5) + (3 \times 1)$$

$$3y + 2 = 20 + 3$$

$$3y + 2 = 23$$

$$3y + 2 - 2 = 23 - 2$$

$$\frac{3y}{3} = \frac{21}{3}$$

$$y = 7$$

y is base seven

3. Given that $203_y = 35_{\text{ten}}$, Find base y

2	0	3	=	35
y^2	y^1	y^0		

$$(2xy^2) + (0xy^1) + (3xy^0) = 35$$

$$2y^2 + 0 + 3 = 35$$

$$2y^2 + 3 - 3 = 35 - 3$$

$$2y^2 + 3 = 35$$

$$2y^2 = 32$$

$$\frac{2y^2}{2} = \frac{32}{2}$$

$$y^2 = 16$$

$$\sqrt{y^2} = \sqrt{16}$$

$$y = 4$$

y is base four

4. Work out the value of the digit represented by m given that $2m1_{\text{four}} = 45_{\text{ten}}$

Solution

$$2m1_{\text{four}} = 45_{\text{ten}}$$

$$(2 \times 4^2) + (m \times 4^1) + (1 \times 4^0) = 45$$

$$(2 \times 16) + (m \times 4) + (1 \times 1) = 45$$

$$32 + 4m + 1 = 45$$

$$33 + 4m = 45$$

$$33 - 33 + 4m = 45 - 33$$

$$4m = 12$$

$$\frac{4m}{4} = \frac{12}{4}$$

$$m = 3$$

Evaluation Activity



1. If $24_h = 40_{\text{five}}$, find base h .
2. Given that $32_h = 17_{\text{ten}}$, find base h .
3. If $23_p = 11_{\text{ten}}$, find the value of p .
4. Given that $101_r = 17_{\text{ten}}$, find base r .
5. Given that $k^2 = 17_{\text{ten}}$, find k .
6. Given that $4p1_{\text{ten}} = 157_{\text{eight}}$, find p .
7. Given that $23_x = 1101_{\text{two}}$, find the value x .
8. Given that $23_k = 15_{\text{ten}}$, find the value of k .
9. If $202_g = 26_{\text{seven}}$, find the value of g .
10. Given that $203_y = 35_{\text{ten}}$, find the value of y .
11. If $24_p = 33_{\text{four}} + 3_{\text{ten}}$, find the base represented by p .

Addition and subtraction of numbers in non-decimal bases

Examples

1. Workout: 1101_{two}

$$\begin{array}{r} 1101_{\text{two}} \\ + 111_{\text{two}} \\ \hline 10100_{\text{two}} \end{array}$$

$$1+1 = 2=2$$

$$1 \text{ rem } 0$$

$$1+1+1 = 3=2$$

$$1 \text{ rem } 1$$

2. Subtract: 101_{two}

$$\begin{array}{r} 101_{\text{two}} \\ - 11_{\text{two}} \\ \hline 10_{\text{two}} \end{array}$$

$$1-1 = 0$$

$$2-1 = 1$$

3. Workout: 333_{five}

$$\begin{array}{r} 333_{\text{five}} \\ - 22_{\text{five}} \\ \hline 311_{\text{five}} \end{array}$$

$$3-2 = 1$$

$$3-2 = 1$$

$$3-0 = 3$$

Evaluation Activity



1. Workout $1010_{\text{two}} + 101_{\text{two}}$
2. Add $104_{\text{five}} + 232_{\text{five}}$
3. Subtract $154_{\text{seven}} - 36_{\text{seven}}$
4. Workout $213_{\text{five}} + 122_{\text{five}}$
5. Add $342_{\text{six}} + 104_{\text{six}}$
6. Workout $11_{\text{two}} + 1_{\text{two}}$
7. Subtract $1110_{\text{two}} - 10_{\text{two}}$
8. Add $203_{\text{four}} + 33_{\text{four}}$
9. Workout $110_{\text{three}} - 11_{\text{three}}$
10. Subtract $204_{\text{five}} - 32_{\text{five}}$
11. The table below shows addition in base five. Complete it correctly.

+	2	3	4
1	3		10
2			
3		11	

Multiplication of numbers in non-decimal bases

Examples

1. Workout: $10_{\text{two}} \times 11_{\text{two}}$

$$\begin{array}{r} 10_{\text{two}} \\ \times 11_{\text{two}} \\ \hline 10_{\text{two}} \\ +100_{\text{two}} \\ \hline 110_{\text{two}} \end{array} \quad \begin{array}{l} 1 \times 0 = 0 \\ 1 \times 1 = 1 \\ 1 \times 0 = 0 \\ 1 \times 1 = 1 \end{array}$$

2. Workout: $102_{\text{five}} \times 13_{\text{five}}$

$$\begin{array}{r} 102_{\text{five}} \\ \times 13_{\text{five}} \\ \hline 311_{\text{five}} \\ +1020_{\text{five}} \\ \hline 1331_{\text{five}} \end{array} \quad \begin{array}{l} 3 \times 2 = 6 \\ 6 - 5 = 1 \text{ rem } 1 \\ 3 \times 0 = 0 \\ 0 \times 1 = 1 \\ 3 \times 1 = 3 \end{array} \quad \begin{array}{l} 1 \times 2 = 2 \\ 1 \times 0 = 0 \\ 1 \times 1 = 1 \end{array}$$

Evaluation Activity



1. Workout $10_{\text{two}} \times 11_{\text{two}}$
2. Workout $104_{\text{five}} \times 32_{\text{five}}$
3. Workout $111_{\text{two}} \times 11_{\text{two}}$
4. Workout $213_{\text{five}} \times 22_{\text{five}}$
5. Workout $42_{\text{six}} \times 14_{\text{six}}$
6. Workout $11_{\text{two}} \times 11_{\text{two}}$
7. Workout $1110_{\text{two}} \times 10_{\text{two}}$
8. Workout $203_{\text{four}} \times 33_{\text{four}}$
9. Workout $110_{\text{three}} \times 11_{\text{three}}$
10. Workout $204_{\text{five}} \times 32_{\text{five}}$

11. The table below shows multiplication in base five. Complete it correctly.

$\times$	2	3	4
1	2		4
2			
3		14	

12. Find the square of 10_{two} .

THEME 2: **NUMERACY**

TOPIC 3: OPERATIONS ON WHOLE NUMBERS

Addition and subtraction of whole numbers

Consider: $4 + 5 = 9$

- In the above mathematical statement, 4 and 5 are addends while 9 is the sum.

Consider: $20 - 13 = 7$

- In the above mathematical statement 20 is the minuend, 13 is the subtrahend and 7 is the difference.

Examples

1. There are 58,167 pens in the box and 789,968 pens in the other box. What is the total number of pens in the two boxes?

Solution

$$\begin{array}{r} 789,968 \\ + 58,167 \\ \hline 848,135 \end{array}$$

2. In a district, 1,238,960 animals were to be vaccinated in two weeks. If 765,874 animals were vaccinated in the first week, how many animals were vaccinated in the second week?

Solution

$$\begin{array}{r} 1,238,960 \\ - 765,874 \\ \hline 473,086 \end{array}$$

3. Workout $345,783 - 472,096 + 432,562$.

Solution

$$\begin{array}{r} 345,783 \\ + 432,562 \\ \hline 778,345 \end{array} \quad \begin{array}{r} 778,345 \\ - 472,096 \\ \hline 306,249 \end{array}$$

Evaluation Activity



1. Find the sum of 231,048 and 524,628.
2. There were 351,272 books in the library and 189,242 more books were added in the same library. how many books are in the library altogether?
3. Find the difference of 678,231 and 354,986
4. What must be added to 237,428 to get 885,643?
5. Subtract 452,367 from 872,291
6. Decrease 845,654 by 518,349.
7. A dairy processed 5,300,450 litres of milk and sold 3,450,833 litres. how many litres were left?



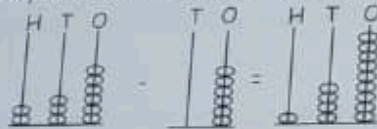
8. Kato is 43 years old, Mike is 15 years younger than him while Aaron is 4 years than Mike. How old is Aaron?

9. Find the missing numbers in the calculations below.

a)
$$\begin{array}{r} 2 \square 56 \\ + \square 483 \\ \hline 119\square9 \end{array}$$

b)
$$\begin{array}{r} 4 \square \\ - 8 \\ \hline 35 \end{array}$$

10. Complete the subtraction statement below.



Multiplication of whole numbers

Consider: $8 \times 6 = 48$

- In the mathematical statement above, 8 is the multiplier, 6 is the multiplicand and 48 is the product.

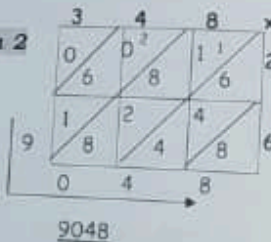
Examples:

1. Find the product of 348 and 26.

Solution 1

$$\begin{array}{r} 348 \\ \times 26 \\ \hline 2088 \\ + 696 \\ \hline 9048 \end{array}$$

Solution 2



2. Kato collects 6,398 eggs on his farm every day, how many eggs does he collect in 74 days?

Solution 1

$$\begin{array}{r} 6398 \\ \times 74 \\ \hline 25592 \\ + 44786 \\ \hline 473452 \end{array}$$

Solution 2



Evaluation Activity

- Find the product of the following:
 - 7,926 and 123
 - 2,508 and 97
 - 847 and 643

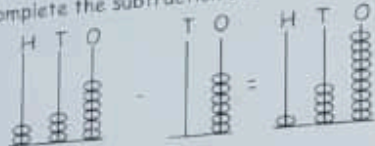
8. Kato is 43 years old, Mike is 15 years younger than him while Aaron is 4 years older than Mike. How old is Aaron?

9. Find the missing numbers in the calculations below.

a)
$$\begin{array}{r} 2 \square 56 \\ + \square 483 \\ \hline 119\square9 \end{array}$$

b)
$$\begin{array}{r} 4 \square \\ - 8 \\ \hline 35 \end{array}$$

10. Complete the subtraction statement below



Multiplication of whole numbers

Consider: $8 \times 6 = 48$

• In the mathematical statement above, 8 is the multiplier, 6 is the multiplicand and 48 is the product.

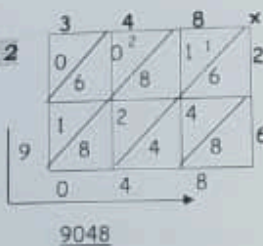
Examples

1. Find the product of 348 and 26.

Solution 1

$$\begin{array}{r} 348 \\ \times 26 \\ \hline 2088 \\ + 696 \\ \hline 9048 \end{array}$$

Solution 2



2. Kato collects 6,398 eggs on his farm every day, how many eggs does he collect in 74 days.

Solution 1

$$\begin{array}{r} 6398 \\ \times 74 \\ \hline 25592 \\ + 44786 \\ \hline 473452 \text{ eggs} \end{array}$$

Solution 2



Evaluation Activity

1. Find the product of the following:

a) 7,926 and 123

b) 2,508 and 97

c) 847 and 643

- 538 suckers are planted in one hectare of land. How many suckers are needed for 32 hectares of land?
- 753 units of water are used for a week in a certain school. How many units of water are used by the school in 45 weeks?
- Workout $3,647 \div 438$.
- A maize mill processes 8,732 bags of flour in a day. How many bags of flour are processed in 128 days?
- Six stores have 500 sacks of maize each. Each sack weighs 115kgs. How many kilogrammes of maize are in the store altogether?
- A match box has 40 match sticks and a carton has 3 dozen of match boxes. How many matchsticks are in a carton?

Division of whole numbers

Consider: $30 \div 2 = 15$

- In the above mathematical statement, 30 is the dividend, 2 is the divisor and 15 is the quotient.

Examples

- On a farm, 14,167 litres of milk are produced in a month of 31 days. How many litres of milk are produced daily?

Solution

$$\begin{array}{r}
 00457 \\
 31 \overline{) 14167} \\
 \underline{31 \times 0 = 0} \\
 14 \\
 \underline{31 \times 0 = 00} \\
 141 \\
 \underline{31 \times 4 = 124} \\
 0176 \\
 \underline{31 \times 5 = 155} \\
 0217 \\
 \underline{7 \times 6 = 42} \\
 000
 \end{array}$$

457 litres are produced daily.

Multiples of the divisor

$$\begin{array}{l}
 1 \times 31 = 31 \\
 2 \times 31 = 62 \\
 3 \times 31 = 93 \\
 4 \times 31 = 124 \\
 5 \times 31 = 155 \\
 6 \times 31 = 186 \\
 7 \times 31 = 217
 \end{array}$$



Evaluation Activity



- Complete the table below correctly

Number of groups	Number of items in each group	Total number of items
12	_____	60
_____	6	24
7	13	_____
8	_____	576

2. Simplify: $5^2 \times 5^3 \times 5$

Solution

Method 1

$$\begin{aligned} 5^2 \times 5^3 \times 5 \\ 5 \times 5 \times 5 \times 5 \times 5 \times 5 \\ 5^6 \end{aligned}$$

Method 2

$$\begin{aligned} 5^2 \times 5^3 \times 5 \\ 5^{(2+3+1)} \\ 5^6 \end{aligned}$$

3. Workout: $4^2 \times 4^3$

Solution 1

$$\begin{aligned} 4^2 \times 4^3 \\ 4 \times 4 \times 4 \times 4 \times 4 \\ 16 \times 16 \times 4 \\ 1024 \end{aligned}$$

Solution 2

$$\begin{aligned} 4^2 \times 4^3 \\ 4^{(2+3)} \\ 4^5 \\ 4 \times 4 \times 4 \times 4 \times 4 \\ 16 \times 16 \times 4 \\ 1024 \end{aligned}$$

4. Evaluate: $9^3 \times 9^{-5}$

Solution

$$\begin{aligned} 9^3 \times 9^{-5} \\ 9 \times 9 \times 9 \times \frac{1}{9 \times 9 \times 9 \times 9 \times 9} \\ \frac{9 \times 9 \times 9}{9 \times 9 \times 9 \times 9 \times 9} \\ \frac{1}{9 \times 9} \\ \frac{1}{81} \end{aligned}$$

Evaluation Activity



1. Simplify the following:

- $p^3 \times p^2$
- $h^4 \times h^3$
- $k^5 \times k^3$
- $b^4 \times b^3 \times b^2$
- $2x^2 \times 3x^3$
- $4p^3 \times 2p^4$

2. Workout the following:

- $8^2 \times 8^{-3}$
- $6^3 \times 6^{-5}$
- $2^4 \times 2^3 \times 2^7$
- $3^{-4} \times 3$

Law of indices in division

Points to note;

- ✓ When dividing powers of the same bases, maintain the base and subtract indices
- ✓ When simplifying indices, give your answer in power form.
- ✓ When working out or evaluating indices, find the value of the expression

Evaluation Activity



1. Work out the following correctly:
a) $48 - 20 + 22$ b) $72 - 4 \times 3$ c) $9 - 9 + 5 - 3$
d) $(8 - 3) - (4 \times 2) + (6 \times 3)$ e) $\frac{3}{10}$ of $60 + \frac{2}{5}$ of 30 f) $452 - 3 \times 9 \times (5 + 6)$
g) $13 \times 36 + 9 + 154$
2. Kiku had 24 fruits and he gave away $\frac{1}{4}$ of them to a friend. Kiku later got 12 fruits more from his daddy. How many fruits did he remain with altogether?
3. Andema had sh. 7500 and received sh. 4500 more from a friend. He later equally shared the money with his elder brother. How much did each get?
4. Simplify: 12 of 8 $(81 - 9) + 18$.
5. Simplify: $\frac{2}{5}$ of $45 + \frac{1}{5}$ of $100 - 8$.
6. Below is an advert of a school on Capital Radio read once a day

The Parent out there, This is the best school you have been looking for. We have a conducive and safe environment, play centres, healthy meals and a quality caring staff. We promote children's talents, morals and confidence. We are completely a Mixed Day school. We are Shine International School, located on Kampala Road. Come experience Education with a difference

Daily charges for an advert read once a day

- ✓ Sh. 10,000 for the first 30 words
- ✓ Sh. 300 for each added word.
(without counting spaces and punctuation marks)

- a) How much does the school pay for the advert per week?
- b) If another school pays sh. 21, 100 daily on the same radio for their advert read once a day, how many words are in that school's advert?

PROPERTIES OF OPERATIONS

- Commutative property
- Associative property
- Distributive property

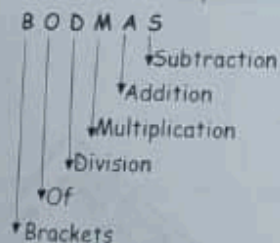
Commutative property

- It states that the order in which numbers are arranged in addition or multiplication does not change the sum or product.
- In commutative property, the order of operation in any two numbers added or multiplied does not change the result.

2. Workout the following:
- | | | |
|---------------------|--------------------|--------------------|
| a) $8,416 \div 4$ | b) $18,372 \div 6$ | c) $7,290 \div 18$ |
| d) $11,776 \div 46$ | e) $6,363 \div 9$ | f) $30,030 \div 3$ |
3. Find the quotient of 5,040 and 63.
4. Share 759 text books amongst 23 schools.
5. A company produces 375,000 pens every day and 250 pens are packed in each box. How many boxes of pens are produced every day?
6. 50,547 people were transported to Luweero gazetted triangle from Buduuda by the government, if 83 chopper trips were used, how many people were transported in each trip?
7. In a box, there are 145 packets of biscuits with 15 pieces of biscuits each packet. One box of biscuits is served to the 87 people who attended a meeting, how many pieces of biscuits did each get?
8. A trader bought 2 boxes of soap each containing 25 bars. He cut 4 pieces from each bar and got sh. 200,000 after selling all the pieces. At what price did he sell each piece?

Combined operations

- In case of more than one operation in a statement, we follow BODMAS.



Examples

2. Simplify: $\frac{1}{2}$ of $50 + 39 \div 3$

Solution

$(\frac{1}{2} \text{ of } 50) + 39 \div 3$ **BODMAS**

$(\frac{1}{2} \times 50) + (39 \div 3)$

$25 + 13$

38

1. Workout: $56 \div (7 \times 8) + 9$

Solution

$56 \div (7 \times 8) + 9$ **BODMAS**

$(56 \div 56) + 9$

$1 + 9$

10

Example:

a) $5+7 = 7+5$

$12 = 12$

b) $6 \times 3 = 3 \times 6$

$18 = 18$

Associative property

- In associative property, the grouping of the given numbers during operation does not change the sum or product
- The alteration or change of brackets in addition or multiplication of more than two numbers does not change the result

Example:

a) $(12+9) + 6 = 12 + (9+6)$

$21+6 = 12+15$

$27 = 27$

b) $5 \times (7 \times 2) = (5 \times 7) \times 2$

$5 \times 14 = 35 \times 2$

$70 = 70$

Distributive property

- In distributive property, the product of the common factor and the sum or difference of any operation is equal to the sum or difference of the product respectively.
- In distributive property, addition or subtraction signs are used outside the brackets while multiplication or division signs are used in the brackets
- To apply distributive property the following steps are key:
 - ✓ Identify the common number in the brackets given
 - ✓ Add or subtract the remaining numbers in brackets.
 - ✓ Multiply or divide the sum or difference of the remaining numbers by the common number.

Examples

1. Workout: $(8 \times 55) + (45 \times 8)$ using the distributive property.

Solution

$(8 \times 55) + (45 \times 8)$

$(55+45)8$

100×8

800

2. Workout: $(4.3 \times 85) - (4.3 \times 55)$ using the distributive property.

$(85-55)4.3$

$= \frac{11}{10} \times 30$

$= 43 \times 3$

$= 129$

3. Workout: $(23+8) \div (19+8)$

$$\begin{aligned} &= (23+19) \div 8 \\ &= 32 \div 8 \\ &= 4 \end{aligned}$$

4) Workout $(83+5) - (33+5)$

$$\begin{aligned} &= (83+33) - 5 \\ &= 50 - 5 \\ &= 10 \end{aligned}$$

Evaluation Activity

1. Fill in the missing values using commutative property

a) $13 \times 9 = \underline{\hspace{1cm}} \times 13$

b) $240 \div \underline{\hspace{1cm}} = 90 \div \underline{\hspace{1cm}}$

c) $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = 36 \times 4$

2. Complete the statements below using associative property

a) $(20 \times \underline{\hspace{1cm}}) \times 7 = 20 \times (\underline{\hspace{1cm}} \times 4)$

b) $85 + (19 + \underline{\hspace{1cm}}) = (\underline{\hspace{1cm}} + 19) + 30$

c) $(17 \times \underline{\hspace{1cm}}) \times 40 = 17 \times (8 \times \underline{\hspace{1cm}})$

3. Workout the following using distributive property

a) $(137 \times 42) - (37 \times 42)$

b) $(17 \times 19) \div (19 \times 13)$

c) $(36 \times 55) \div (36 \times 45)$

d) $(48 \times 74) - (48 \times 54)$

e) $(78 \div 4) \div (22 \div 4)$

f) $(123 \div 6) - (3 \div 6)$

h) $(56 \div 10) \div (14 \div 10)$

i) $(56 \div 10) - (16 \div 6)$

j) $(\frac{2}{3} \text{ of } 180) - (120 \text{ of } \frac{2}{3})$

INDICES (EXPONENTS)

- An index or exponent shows the number of times a letter or number base is multiplied by its self.

Study the table below

Power	Expanded form
5^3	$5 \times 5 \times 5$
4^5	$4 \times 4 \times 4 \times 4 \times 4$
p^3	$p \times p \times p$
7^2	$\frac{1}{2} \times \frac{1}{2}$
k^4	$k \times k \times k \times k$
y^3	$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$
2^1	2
9^0	1
X^0	1

Note: Any non zero number raised to **zero** is one (1) and any non zero digit raised to one is that very number.



Expressing numbers in terms of powers

- Therefore m^4 is a power where m is the base and 4 is the index showing the number of times the base is to be multiplied.

1. Express 16 in powers of 2.

$$16 = 2 \times 2 \times 2 \times 2$$

$$16 = 2^4$$

2	16
2	8
2	4
2	2
	1

2. Express 625 in powers of 5

$$625 = 5 \times 5 \times 5 \times 5$$

$$625 = 5^4$$

5	625
5	125
5	25
5	5
	1

Evaluation Activity



- Express 27 in powers of 3
- Write 125 in powers of 5
- Express 16 in powers of 4
- Write 81 using powers of 3

- Express 64 in powers of 4
- Write 32 in powers of 2
- Express 128 in powers of 2

Addition and subtraction of simple indices

Examples

- Simplify: $6^2 + 3^3$

Solution;

$$6^2 + 3^3$$

$$6 \times 6 + (3 \times 3 \times 3)$$

$$36 + 27$$

$$63$$

- Evaluate $5^2 + 5^{-3}$

Solution;

$$5^2 + 5^{-3}$$

$$5 \times 5 \times \left(\frac{1}{5} \times \frac{1}{5} \times \frac{1}{5}\right)$$

$$\frac{5}{1} \times \frac{5}{1} \times \left(\frac{1}{5} \times \frac{1}{5} \times \frac{1}{5}\right)$$

$$\frac{1}{5}$$

Evaluation Activity



Simplify the following.

- $3^2 + 2^0 + 4^1$
- $5^2 - 2^4$
- $3^3 + 2^3$
- $6^2 + 6^0 + (2+1)^2$
- $(2^2-1)^4 - 6^2 - 4$
- $7^3 - (3^3 + 2^3)$
- $8^{-1} + \frac{1}{8}$
- $5^{-3} \times 2$

Law of indices in multiplication

Points to note:

- ✓ When multiplying powers of the same base, maintain the base and add the powers.
- ✓ When simplifying indices, give your answer in power form.
- ✓ When working out or evaluating indices, find the value of the expression.

Examples

- Simplify: $n^2 \times n$

Solution 1

$$n^2 \times n$$

$$n^2 \times n$$

$$n \times n \times n$$

$$n^3$$

Solution 2 (using the law of indices)

$$n^2 \times n^1$$

$$n^{(2+1)}$$

$$n^3$$

Evaluation Activity

- Express the following numbers in standard form.
 - 2300
 - 40035
 - 368000
 - 456.98
- Write the following numbers in standard form.
 - 0.056
 - 0.00056
 - 0.0304
 - 0.00678
- Work out the following operations
 - $(2.4 \times 10^3) \times (2 \times 10^3)$
 - $(42 \times 10^6) - (1.4 \times 10^4)$
 - $(5.5 \times 10^4) \times (2 \times 10^3)$
 - $(1.44 \times 10^6) - (6 \times 10^3)$

Finding the number which has been expressed in standard form/ scientific notation

Examples

- Express 3.08×10^3 as a single number.

Solution

$$3.08 \times 10^3$$

$$308 \times 10 \times 10$$

$$30800$$

$$3080$$

- What number has been expressed in standard form to give 2.3×10^{-4} ?

Solution

$$2.3 \times 10^{-4}$$

$$\frac{2.3}{10 \times 10 \times 10 \times 10}$$

$$\frac{2.3}{10000}$$

$$0.00023$$

Evaluation Activity

- Find the number whose standard form is:
 - 3.4×10^2
 - 6.45×10^4
 - 9.203×10^3
 - 7.23×10^1
- Find the number whose standard form is:
 - 3.8×10^{-3}
 - 2.03×10^{-2}
 - 4.28×10^{-4}
 - 3.64×10^{-1}

PRIME FACTORIZATION OF NUMBERS

- Prime factorization is a way of finding prime factors of a number.
- Prime factors are prime numbers which exactly divide a given number. (2, 3, 5, 7, 11, 13, ...)
- Prime factors of a number are written in three different ways:
 - Set notation/ subscript form.
 - Power form/ exponential form.
 - Product/ multiplication form.

Examples

- Prime factorize 18.

Ladder method

2	18
3	9
3	3
	1

Set notation/ subscript form

$$F18 = \{2, 3, 3\}$$

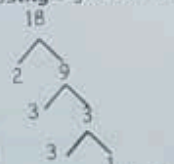
Power form

$$F18 = 2^1 \times 3^2$$

Product form

$$F18 = 2 \times 3 \times 3$$

Using a factor tree



Set notation/ subscript form

$$F18 = \{2, 3, 3\}$$

Power form

$$F18 = 2^1 \times 3^2$$

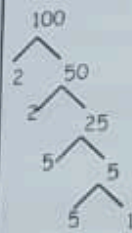
Product form

$$F18 = 2 \times 3 \times 3$$

Solution

2	100
2	50
5	25
5	5
	1

Factor tree



$$F_{100} = \{2_1, 2_2, 5_1, 5_2\}$$

- ### Finding prime factorized numbers

$$= 180$$

- The value of m in the Venn diagram is
- 
- The Venn diagram shows two overlapping sets, F_{12} and F_{15} . The elements of F_{12} are 2_1 and 2_2 . The elements of F_{15} are 5_1 and m . The intersection of the two sets contains the element m .

Representing prime factors on Venn diagrams

Examples

1. a) Represent the prime factors of 12 and 18 on a Venn diagram

2	12
2	6
3	3
	1

2	18
3	9
3	3
	1



- b) Find the GCF of 12 and 18.

$$\begin{aligned} \text{G.C.F} &= \text{product of } F_{12} \cap F_{18} \\ F_{12} \cap F_{18} &= (2_1, 3_1) \\ \text{Therefore, G.C.F} &= 2 \times 3 \\ &= 6 \end{aligned}$$

- c) Work out the LCM of 12 and 18.

$$\begin{aligned} \text{Product of } F_{12} \cup F_{18} \\ F_{12} \cup F_{18} &= (2_1, 2_2, 3_1, 3_2) \\ &= 2 \times 2 \times 3 \times 3 \\ &= 4 \times 9 \\ &= 36 \end{aligned}$$

Evaluation Activity

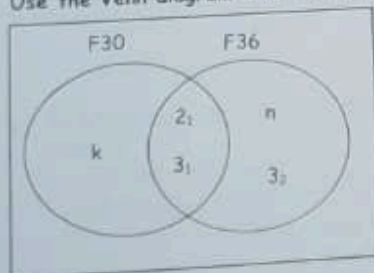


- Prime factorize 90 and 60
 - Show the prime factors of 90 and 60 on a Venn diagram
 - Calculate the GCF of 90 and 60
 - Work out the LCM of 90 and 60
- Given the numerals 30 and 40
 - Show the prime factors of 30 and 70 on a Venn diagram
 - Work out the GCF of 30 and 70
 - Calculate the LCM of 30 and 70
- Prime factorize 48 and 72
 - Represent the prime factors of 48 and 72 on a Venn diagram
 - Calculate the GCF of 48 and 72
 - Find the LCM of 48 and 72



Application of prime factors on a Venn diagram

1. Use the Venn diagram below to answer the questions that follow



- a) Find the value of k.

$$\begin{aligned} \text{Solution 1} \\ k \times 2 \times 3 &= 30 \\ 6k &= 30 \\ \frac{6k}{6} &= \frac{30}{6} \\ k &= 5 \end{aligned}$$

Solution 2

$$\begin{array}{r} 30 \\ 2 \overline{) 30} \\ \underline{20} \\ 10 \\ 5 \overline{) 10} \\ \underline{5} \\ 5 \\ 5 \overline{) 5} \\ \underline{5} \\ 0 \end{array}$$

(2, 3, 5)
k = 5

Evaluation Activity



Solve for the following unknowns

- $2^x = 32$
- $3^x = 27$
- $5^x = 625$
- $6^x + 1 = 217$
- $5^x = 25$



Application of indices (part 2)

- Here we apply the laws of indices in multiplication or division to solve for the unknowns.
- We also express the given numbers as powers of the given base in the equation through factorization.

Examples

3. Solve for p if

$$3^p \times 3^2 = 81$$

$$3^p \times 3^2 = 3^4$$

$$p + 2 = 4$$

$$p + 2 - 2 = 4 - 2$$

$$p = 2$$

3	81
3	27
3	9
3	3
	1

2. Find r if $5^r \div 5^3 = 25$

$$5^r \div 5^3 = 5^2$$

$$r - 3 = 2$$

$$r - 3 + 3 = 2 + 3$$

$$r = 5$$

5	25
5	5
	1

1. Solve for a if

$$3^a \div 81 = 1$$

$$3^a \div 81 = 1$$

$$3^a \div 3^4 = 3^0$$

$$a - 4 = 0$$

$$a - 4 + 4 = 0 + 4$$

$$a = 4$$

3	81
3	27
3	9
3	3
	1

Evaluation Activity



Solve for the following unknowns:

a) $3^p \times 3^2 = 27$

b) $5^r \div 5^2 = 125$

c) $2^x \times 2^2 = 64$

d) $3^x \div 3^3 = 27$

e) $5^d \times 5^2 = 125$

f) $5^x \div 125 = 1$

g) $3^a \times 27 = 81$

h) $2^x \div 64 = 1$

i) $2^{x+1} \div 2^5 = 1$

Application of indices (part 3)

Examples

1. Solve for p in $5^{p+1} \times 2^3 = 200$

$$5^{p+1} \times 2^3 = 200$$

$$5^{p+1} \times 2 \times 2 \times 2 = 2 \times 2 \times 2 \times 5 \times 5$$

$$5^{p+1} \times \cancel{2} \times \cancel{2} \times \cancel{2} = \cancel{2} \times \cancel{2} \times \cancel{2} \times 5 \times 5$$

$$\cancel{2} \times \cancel{2} \times \cancel{2} \quad \cancel{2} \times \cancel{2} \times \cancel{2}$$

$$5^{p+1} = 5^2$$

$$p + 1 = 2$$

$$p + 1 - 1 = 2 - 1$$

$$p = 1$$

2	200
2	100
2	50
5	25
5	5
	1

2. Solve for p: $2^p \times 3^2 = 72$

Solution

$$2^p \times 3^2 = 72$$

$$2^p \times 3^2 = 2 \times 2 \times 2 \times 3 \times 3$$

$$2^p \times 3^2 = 2^3 \times 3^2$$

$$2^p = 2^3$$

$$p = 3$$

b) Work out the value of n .

Solution

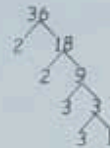
$$n \times 3 \times 3 \times 2 = 36$$

$$18n = 36$$

$$\frac{18n}{18} = \frac{36}{18}$$

$$n = 2$$

$(2_1, 2_2, 3_1, 3_2)$



c) Find the LCM of 30 and 36

$$\text{LCM} = 2 \times 3 \times 2 \times 3 \times 5$$

$$= 6 \times 6 \times 5$$

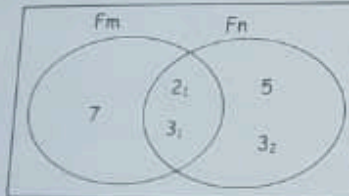
$$= 36 \times 5$$

$$= 180$$

Evaluation Activity



1. Use the Venn diagram below to answer the questions that follow.



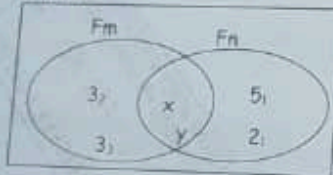
a) Find the value of:

i) m

ii) n

b) Work out the LCM and GCF of m and n .

2. Study the Venn diagram below carefully and use it to answer the questions that follow.



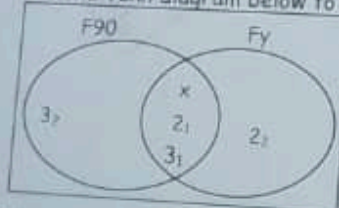
a) Find the value of:

iii) x

iv) y

b) Work out the LCM and GCF of m and n .

3. Use the Venn diagram below to answer the questions that follow.



a) Find the value of:

i) x

ii) y

b) Work out the LCM and GCF of x and y .

4. The intersection set for two numbers A and B is $\{2_1, 3_1\}$. If $A - B = \{2_2, 3_2\}$ and $B - A = \{5_1\}$.

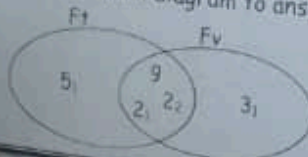
a) Find the value of A

b) Find the value of B

c) Find the LCM of A and B

d) Work out the GCF of A and B

5. Use the Venn diagram to answer the questions that follow.



If the LCM of t and v is 780.

a) Find the value of g .

b) Calculate the GCF of t and v .

c) Find the value of t and v .

Evaluation Activity



Find the unknown indices in the following:

a) $2^a \times 3^3 = 108$

b) $2^2 \times 5^a = 100$

i) $(2^{3x}) - (3^{3x}) = 64$

c) $2^a \times 3^2 = 36$

d) $5^x \times 2^3 = 200$

j) $3^{x-1} \times 3^{2x-1} = 3^9$

e) $3^d \times 5^2 = 75$

f) $3^2 \times 5^y = 225$

STANDARD FORM/SCIENTIFIC NOTATION

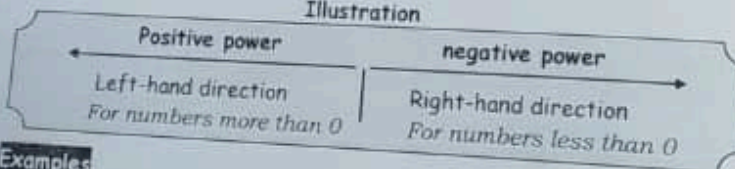
- A number in standard form or scientific notation is written as $P \times 10^n$ where P is a counting number from 1 to 9 and n is a positive or negative integer or zero (0).
- The standard form gives one digit in the place value of whole numbers before a decimal point and these numbers are: 1, 2, 3, 4, 5, 6, 7, 8 and 9.
- The new decimal fraction is multiplied by the power of ten where the index shows the number of steps the decimal point has been moved to its right position.

Writing numbers in standard form/ scientific notation

Note:

- When a decimal point is moved to the left-hand direction, a positive power is used.
- When a decimal point is moved to the right-hand direction, a negative power is used.

Illustration



Examples

1. Express 4763 in scientific notation.

Solution

$$\overline{4763}.$$

$$4763 \div 10 = 476.3$$

$$476.3 \div 10 = 47.63$$

$$47.63 \div 10 = 4.763$$

$$4763 = 4.763 \times 10^3$$

2. Write 832.76 in standard form.

Solution

$$\overline{832.76}$$

$$832.76 \div 10 = 83.276$$

$$83.276 \div 10 = 8.3276$$

$$832.76 = 8.3276 \times 10^2$$

3. Write 30600 in standard form.

Solution

$$\overline{30600}.$$

$$30600 \div 10 = 3060$$

$$3060 \div 10 = 306$$

$$306 \div 10 = 30.6$$

$$30.6 \div 10 = 3.06$$

$$3.06 \times 10^4$$

The zeros after 6 are not significant.

4. Express 0.000724 in scientific notation.

Solution

$$\overline{0.000724}$$

$$0.000724 \times 10 = 0.00724$$

$$0.00724 \times 10 = 0.0724$$

$$0.0724 \times 10 = 0.724$$

$$7.24 \times 10^{-4}$$

Application of lowest common multiple (LCM)

Examples

1. What least number of oranges can be shared equally among 16 or 18 pupils without leaving a remainder?

Solution

2	16	18
2	8	9
2	4	9
2	2	9
3	1	9
3	1	3
	1	1

$$\begin{aligned} \text{LCM} &= 2 \times 2 \times 2 \times 3 \times 3 \\ &= 4 \times 4 \times 9 \\ &= 144 \text{ oranges} \end{aligned}$$



2. Find the least number of sweets when shared among a group of 12 or 15 girls leaves seven as a remainder.

Solution

2	12	15
2	6	15
3	3	15
5	1	5
	1	1

$$\begin{aligned} \text{LCM} &= 2 \times 2 \times 3 \times 5 \\ &= 4 \times 15 \\ &= 60 \\ \text{Number} &= \text{LCM} + \text{remainder} \\ &= 60 + 7 \\ &= 67 \text{ sweets} \end{aligned}$$

3. Two bells one for upper primary and one for lower primary at Edger parents' school ring at an interval of 30 minutes and 40 minutes respectively. If they first ring together at 8:00 am;

- a) After how many minutes will they ring together again?

Solution

LCM

2	30	40
2	15	20
2	15	10
3	15	5
5	5	5
	1	1

$$\begin{aligned} \text{LCM} &= 2 \times 2 \times 2 \times 3 \times 5 \\ &= 4 \times 6 \times 5 \\ &= 24 \times 5 \\ &= 120 \text{ minutes} \\ \text{They will ring together after 120 minutes} \end{aligned}$$

- b) At what time will ring together again?

Solution

$$\begin{aligned} 60 \text{ minutes} &= 1 \text{ hour} \\ 120 \text{ minutes} &= (120 - 60) \text{ hours} \\ &= 2 \text{ hours} \end{aligned}$$

Time when to ring together

Hrs	:	Mins
8	:	00
2	:	00
Time 10	:	00 am

Evaluation Activity



- Express the following numbers in standard form
 - 2300
 - 40035
 - 368000
 - 456.98
- Write the following numbers in standard form
 - 0.056
 - 0.00056
 - 0.0304
 - 0.00678
- Work out the following operations:
 - $(2.4 \times 10^3) \times (2 \times 10^3)$
 - $(5.5 \times 10^4) \times (2 \times 10^3)$
 - $(42 \times 10^6) - (14 \times 10^5)$
 - $(1.44 \times 10^6) - (6 \times 10^3)$

Finding the number which has been expressed in standard form/ scientific notation

Examples

- Express 3.08×10^3 as a single number

Solution:

$$\begin{array}{r} 3.08 \times 10^3 \\ 308 \times 10 \times 10 \times 10 \\ \hline 3080 \end{array}$$

- What number has been expressed in standard form to give 2.3×10^{-4}

Solution:

$$\begin{array}{r} 2.3 \times 10 \\ \frac{23}{10} \times \frac{1}{10} \times \frac{1}{10} \times \frac{1}{10} \\ \hline 0.00023 \end{array}$$

Evaluation Activity



- Find the number whose standard form is:
 - 3.4×10^2
 - 6.45×10^4
 - 9.203×10^3
 - 7.23×10^1
- Find the number whose standard form is:
 - 3.8×10^{-3}
 - 2.03×10^{-2}
 - 4.28×10^{-4}
 - 3.64×10^{-1}

PRIME FACTORIZATION OF NUMBERS

- Prime factorization is a way of finding prime factors of a number.
- Prime factors are prime numbers which exactly divide a given number. {2, 3, 5, 7, 11, 13}
- Prime factors of a number are written in three different ways:
 - Set notation/ subscript form
 - Power form/ exponential form
 - Product/ multiplication form

Examples

- Prime factorize 18.

Ladder method

2	18
3	9
3	3
	1

Set notation/ subscript form

$$F18 = \{2, 3, 3\}$$

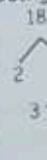
Power form

$$F18 = 2^1 \times 3^2$$

Product form

$$F18 = 2 \times 3 \times 3$$

Using a factor tree



Set notation/ subscript form

$$F18 = \{2, 3, 3\}$$

Power form

$$F18 = 2 \times 3^2$$

Product form

$$F18 = 2 \times 3 \times 3$$

Evaluation Activity



- Find the LCM of the following.
 - 36 and 24
 - 14 and 21
 - 20 and 25
- What number can be divided by 16 and 12 without leaving a remainder?
- What least number of sweets that can be shared equally among 7 or 8 children leaving 3 as a remainder?
- In Shine Junior School, there are three bells one for nursery, one for lower primary and the other for upper primary ring at an interval of 20 minutes, 30 minutes and 40 minutes respectively. If they first ring together at 9:15 a.m.
 - After how many minutes will they ring together again?
 - At what time will they ring together again?
- The teacher had a certain number of pens. When he shared among them 24 pupils 36 pens remained. How many pens did the teacher have?
- In a bus park, three buses A, B and C set off for their journeys at intervals of 15 minutes, 50 minutes and 30 minutes respectively.
 - After how many hours will all the buses set off together?
 - If they first set off together at 6:45 am, at what time of the day do they set off together for the second time?

Application of GCF

Examples

- Find the highest number of children that can equally share 12 sweets and 8 pencils.

2	12	8
2	6	4
3	2	

$$\begin{aligned} \text{GCF} &= 2 \times 2 \\ &= 4 \text{ children} \end{aligned}$$

- What is the highest number of storage houses that can keep 24 potato sacks and 18 rice bags distributed?

2	24	18
3	12	9
3	4	3

$$\begin{aligned} \text{GCF} &= 2 \times 3 \\ &= 6 \text{ storage houses} \end{aligned}$$

Evaluation Activity



- Find the greatest number that can exactly divide 54 or 108.
- What highest number of schools in a district can evenly share 180 textbooks and 120 desks?
- The head teacher of Shine International School gave out 24 certificates, 18 laptops and 30 school bags as gifts at the end of year Thanksgiving Day.
 - If the gifts were shared equally, find the number of children that shared.
 - How many laptops did each child get?
 - How many certificates did each child get?

4. Janet sells 21 pairs of shoes and 28 pairs of stockings to some customers a week. If the customers buy the same number of pairs of shoes and stockings.
- What is the highest number of customers she sells to?
 - How many pairs of shoes does each of them buy?

Application of LCM and GCF (Part 1)

Note:

✓ Product of two numbers = Product of their LCM and GCF

Examples

1. Find the product of two numbers whose LCM and GCF are 36 and 6 respectively.

Solution

$$\begin{aligned}\text{Product of the two numbers} &= \text{product of LCM and GCF} \\ &= 36 \times 6 \\ &= 108\end{aligned}$$

2. Two numbers have a product of 112. Calculate the GCF of the two numbers if their LCM is 28.

Solution

$$\begin{aligned}\text{Let the GCF be } k \\ k \times 28 &= 112\end{aligned}$$

$$\begin{aligned}\frac{28k}{28} &= \frac{112}{28} \\ k &= 4\end{aligned}$$

3. The LCM of two numbers is 24 and Find their GCF is 4. Find the second number if the first number is 8.

$$\begin{aligned}\text{Second number} &= \frac{\text{LCM} \times \text{GCF}}{\text{Given number}} \\ \text{Second number} &= \frac{24 \times 4}{8}\end{aligned}$$

$$\begin{aligned}\text{Second number} &= 3 \times 4 \\ \text{Second number} &= 12\end{aligned}$$

Evaluation Activity



- Find the product of k and m if their GCF and LCM is 42 and 3 respectively.
- The LCM of two numbers is 24 and their GCF is 4. Find the second number if the first number is 8.
- Given the two numbers 12 and m , find the value of m if their GCF and LCM is 6 and 36 respectively.
- The product of two numbers whose LCM is 48 is 288. Find their GCF.
- The LCM of x and y is 120, their GCF is 10. Find the value of y if x is four times the GCF.

Application of LCM and GCF (Part 2)

Note:

$$1^{\text{st}} \text{ Ratio} \times 2^{\text{nd}} \text{ ratio} \times \text{GCF} = \text{LCM}$$

Examples

1. The ratio of two numbers is 3:4. If their GCF is 2,

- a) Find their LCM

Solution

$$1^{\text{st}} \text{ ratio} \times 2^{\text{nd}} \text{ Ratio} \times \text{GCF} = \text{LCM}$$

$$3 \times 4 \times 2 = \text{LCM}$$

$$12 \times 2 = \text{LCM}$$

$$24 = \text{LCM}$$

$$\therefore \text{LCM} = 24$$

- b) Find each of the numbers.

Method 1 (Using GCF)

Smaller no.	Bigger no.
Smaller ratio $\times$ GCF	Bigger no. $\times$ GCF
3×2	4×2
6	8

Method 2. (Using LCM)

Smaller no.	Bigger no.
$= \frac{\text{LCM}}{\text{Bigger ratio}}$	$= \frac{\text{LCM}}{\text{Smaller ratio}}$
$= \frac{24}{4}$	$= \frac{24}{3}$
$= 6$	$= 8$

2. The ratio of two numbers is 4:3. Their LCM is 36.

- a) Find their GCF.

Solution

Let the GCF be k .

$$1^{\text{st}} \text{ ratio} \times 2^{\text{nd}} \text{ ratio} \times \text{GCF} = \text{LCM}$$

$$4 \times 3 \times k = 36$$

$$12k = 36$$

$$\frac{12k}{12} = \frac{36}{12}$$

$$k = 3$$

- b) Find each of the numbers.

Solution

$$\text{Smaller no.} = \text{Smaller ratio} \times \text{GCF}$$

$$= 3 \times 3$$

$$= 9$$

$$\text{Bigger no.} = \text{Bigger ratio} \times \text{GCF}$$

$$= 4 \times 3$$

$$= 12$$

Evaluation Activity



- The ratio of two numbers is 2:3. Their GCF is 4.
 - Find their LCM.
 - Work out each of the numbers.
- The ratio of two numbers is 5:3. Their LCM is 150.
 - Find their GCF.
 - Find the smaller number.
- The ratio of two numbers is $x : 7$ and their GCF is 4. If their LCM is 84, find the value of x .

SQUARES AND SQUARE ROOTS OF WHOLE NUMBERS

- A square number is got by multiplying a counting number by its self.
- A square root is a number that is multiplied by its self to give a square number.
- In $6 \times 6 = 36$, 6 is the square root of 36 and 36 is the square of 6.

Finding squares and square roots of whole numbers

1. Find the square of:

a) 7

$$7^2 = 7 \times 7 \\ = 49$$

b) 13

$$13^2 = 13 \times 13 \\ = 169$$

c) 18

$$18^2 = 18 \times 18 \\ = 324$$

2. Finding square roots of whole numbers

a) 36

$$\sqrt{36} = \sqrt{(2 \times 2) \times (2 \times 2)} \\ \sqrt{36} = 2 \times 2 \\ \sqrt{36} = 6$$

2	36
2	18
3	9
3	3
	1

b) 196

$$\sqrt{196} = \sqrt{2 \times 2 \times 7 \times 7} \\ = 2 \times 7 \\ \sqrt{196} = 14$$

2	196
2	98
7	49
7	7
	1

3. Find the square root of 100

$$\sqrt{100} = \sqrt{(2 \times 2) \times (5 \times 5)} \\ = 2 \times 5 \\ \sqrt{100} = 10$$

2	100
2	50
5	25
5	5
	1

Evaluation Activity



1. Work out the square of the following:

a) 11

c) 19

e) 47

b) 15

d) 25

f) 114



2. Calculate the square root of the following.
- | | | |
|--------|--------|---------|
| a) 64 | c) 144 | e) 625 |
| b) 121 | d) 324 | f) 1600 |
3. Calculate the area of a square whose side measures:
- | | | |
|---------|--------|----------|
| a) 14cm | c) 21m | e) 41dm |
| b) 17cm | d) 22m | f) 103dm |
4. Evaluate the following.
- | | |
|-----------------|--------------------------|
| a) 20^2 | c) $\sqrt{63 \cdot 7^0}$ |
| b) $\sqrt{196}$ | d) |

Finding squares and square roots of fractions

Examples

1. Find the square of the following numbers.

a) $\frac{2}{3}$

$$\left(\frac{2}{3}\right)^2 = \frac{2}{3} \times \frac{2}{3}$$

$$= \frac{4}{9}$$

b) $2\frac{2}{3}$

$$\left(2\frac{2}{3}\right)^2 = \left(\frac{8}{3}\right)^2$$

$$= \frac{8}{3} \times \frac{8}{3}$$

$$= \frac{64}{9}$$

$$= 7\frac{1}{9}$$

c) 2.5

$$(2.5)^2 = \left(\frac{25}{10}\right)^2$$

$$= \frac{25}{10} \times \frac{25}{10}$$

$$= \frac{625}{100}$$

$$= 6.25$$

2. Finding the square roots of fractions

a) $\frac{9}{16}$

$$\frac{9}{16} = \sqrt{\frac{9}{16}}$$

$$= \frac{\sqrt{9}}{\sqrt{16}}$$

$$= \frac{3}{4}$$

b) $\sqrt{12\frac{1}{4}}$

$$= \sqrt{\frac{49}{4}}$$

$$= \frac{\sqrt{49}}{\sqrt{4}}$$

$$= \frac{7}{2}$$

$$= 3\frac{1}{2}$$

c) 2.25

$$2.25 = \sqrt{2.25}$$

$$= \sqrt{\frac{225}{100}}$$

$$= \frac{\sqrt{225}}{\sqrt{100}}$$

$$= \frac{15}{10}$$

$$= 1.5$$

3. Evaluate the following

a) $\left(\frac{5}{6}\right)^2$

c) $\sqrt{\frac{25}{49}}$

b) $(0.7)^2$

d) $\sqrt{0.36}$

Evaluation Activity



1. Find the square of the following:

a) $\frac{5}{7}$

b) $\frac{14}{17}$

c) $3\frac{4}{9}$

d) $2\frac{1}{5}$

e) 0.3

f) 1.2

g) 2.4

h) 1.4

2. Find the square roots of the following:

- a) 169 b) 0.81 c) $\frac{16}{25}$
 d) $12\frac{1}{4}$ e) 2.25 f) $\frac{64}{121}$

Application of square roots

Examples

1. Find the value of p in $p^2 = 36$.

Solution:

$$p^2 = 36$$

$$\sqrt{p \times p} = \sqrt{(2 \times 2) \times (3 \times 3)}$$

$$p = 2 \times 3$$

$$p = 6$$



2. The area of a square room is 400m^2 . Calculate the length of its side.

$$\text{Side} \times \text{side} = \text{area}$$

$$S^2 = 400\text{m}^2$$

$$\sqrt{S^2} = \sqrt{400}$$

$$S = 20\text{m}$$

Evaluation Activity



1. Solve for the unknowns:

a) $r^2 = 196$

c) $q^2 = 361$

b) $k^2 = 256$

d) $n^2 = 441$

2. Calculate the length of the side of a square whose area is:

a) 64 cm^2

c) 576dm^2

b) 900m^2

d) 289cm^2

3. Given that $a=3$ and $b=4$. Find the value of c if $c^2 = a^2 + b^2$.

4. Find the value of p if $(p \times p) = 81$

5. The area of a square flower garden is $5\frac{1}{9}\text{m}^2$. Find length of its sides.

Cube root of numbers

- A cube number is a number got by multiplying a counting number by its self thrice.
- A number multiplied by its self thrice to give a cube number is called a cube root.
- In $2 \times 2 \times 2 = 8$, 2 is a cube root of eight as 8 is a cube number.
- The symbol of a cube root is $\sqrt[3]{}$



Finding cube roots

Examples

1. Find the cube root of 64.

Solution:

$$\sqrt[3]{64}$$

$$= \sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2)}$$

$$= 2 \times 2$$

$$= 4$$

2	64
2	32
2	16
2	8
2	4
2	2
	1

2. Solve for r : $r^3 = 125$

$$r^3 = 125$$

$$\sqrt[3]{r^3} = \sqrt[3]{125}$$

$$r = \sqrt[3]{(5 \times 5 \times 5)}$$

$$r = 5$$

5
5
5

2. Calculate the length of the side of a cube whose volume is 27cm^3

$$S \times S \times S = V$$

$$S^3 = 27\text{cm}^3$$

$$\sqrt[3]{S} = \sqrt[3]{27}$$

$$= \sqrt[3]{(3 \times 3 \times 3)}$$

$$S = 3\text{ cm}$$

3	27
3	9
3	3
	1

Evaluation Activity



- Given that $p^3 = 729$. Find the value of p .
- Work out the cube root of:
 - 343
 - 512
- Calculate the length of the side of a cube whose volume is:
 - 1000cm^3
 - 64dm^3
 - 216m^3
 - 125m^3
- Find the length of a cube whose capacity is 2.7 litres.

THEME 2: **NUMERACY**

TOPIC 4: PATTERNS AND SEQUENCES

Divisibility tests

- Divisibility test is the quicker way of finding out whether a number is exactly divisible by another number or not

Divisibility test for 2, 5 and 10

Divisibility test for 2

- ❖ A number is exactly divisible by two when it is even i.e. when its last digit is 0, 2, 4, 6 or 8 e.g. 348, 792, 120, 364, 138, 1506, etc.

Divisibility test for 5

- ❖ A number is exactly divisible by five when its last digit is either zero or five e.g. 5, 10, 15, 20, 25, 30, 35, etc.

Divisibility test for 10

- ❖ A number is exactly divisible by ten when its last digit is zero e.g. 10, 100, 310, 5700 etc.

Problem Solving Activity



1. Underline the numbers which are exactly divisible by 2
a) 9070 b) 8629 c) 274 d) 1,538 e) 6,483.
2. Tick all the numbers that are exactly divisible by five from the list below
443, 655, 7,695, 30,003 and 100,010
3. Circle the numbers that are exactly divisible by ten in the list below
601, 450, 10,001, 10,000, and 40,401

Divisibility test for 3, 6 and 9

Divisibility test for 3

- ❖ A number is exactly divisible by 3 when the sum of its digit is a multiple of three

Example:

Which of the following numbers 234 and 632 is exactly divisible by three?

234

$$\begin{aligned}\text{Sum of digits} &= 2 + 3 + 4 \\ &= 9\end{aligned}$$

9 is a multiple of three.

234 is exactly divisible by 3.

632

$$\begin{aligned}\text{Sum of digits} &= 6 + 3 + 2 \\ &= 11\end{aligned}$$

11 is not a multiple of three.

632 is not exactly divisible by three

Divisibility test for 6

- ❖ A number is exactly divisible by six when it is divisible by 2 and 3.
- ❖ A number is exactly divisible by 6 if it is an even number and when the sum of all its digits is a multiple of 3.

Examples:

Without dividing determine whether the following numbers are exactly divisible by six.

a) 2,346

The number is even.

$$\begin{aligned}\text{Sum of the digits} &= 2 + 3 + 4 + 6 \\ &= 15\end{aligned}$$

15 is a multiple of three.

Therefore, 2346 is exactly divisible by six.

b) 23,517

The number is odd, not divisible by six.

c) 14,536

The number is even

$$\text{Sum of the digits } (1 + 4 + 5 + 3 + 6) = 19$$

19 is not a multiple of 3. Therefore, 14,536 is not divisible by six.

Divisibility test for 9

♦ A number is exactly divisible by nine when the sum of its digits is a multiple of 9.
e.g. 9, 18, 27, 36, 45, 54, 63, 72, 81, ...

Examples:

Without dividing determine whether the following numbers are exactly divisible by nine.

a) 234

$$\begin{aligned}\text{Sum of the digits} &= 2 + 3 + 4 \\ &= 9\end{aligned}$$

9 is a multiple of nine.

Therefore, 234 is exactly divisible by 9.

b) 56,328

$$\begin{aligned}\text{Sum of the digits} &= 5 + 6 + 3 + 2 + 8 \\ &= 24\end{aligned}$$

24 is not a multiple of 9.

Therefore, 56,328 is not exactly divisible by nine.

Evaluation Activity



- Which of the following numbers are exactly divisible by 3.
a) 32,454 b) 32,145 c) 12,134 d) 2,754
- Without dividing, find out which numbers are exactly divisible by six from 334, 4536 and 4,578.
- Which of the following numbers are exactly divisible by nine?
a) 2709 b) 6,145 c) 45,810 d) 321,456
- Identify all the numbers which are exactly divisible by three in the list below.
11, 33, 123, 1243, 873 and 4320

5. Without dividing identify the numbers which are exactly divisible by three in the list below 237, 762, 651 and 4560.
6. Given that $7p6$ is exactly divisible by 3, find the value of p .

Divisibility for 4 and 8

Divisibility test for 4

✧ A number is exactly divisible by four when the number formed by its last two digits is a multiple of 4 (divisible by 4). e.g 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56.

✧ A number is exactly divisible by 4 if it ends with two zeros (00).

Examples:

Without dividing, find out whether the numbers below are exactly divisible by four.

a) 112

Last two digits form 12

12 is a multiple of 4

112 exactly divisible by 4

b) 7,218

Last two digits form 18

18 is not a multiple of 4

7,218 is not exactly divisible by 4

Divisibility test for 8

✧ A number is exactly divisible by 8 if the number formed by its last three digits is a multiple of 8 (divisible by 8).

Examples:

1. Without dividing, show whether 7684 is exactly divisible by 8.

Solution

$7\overline{)684}$

$\underline{684} = 85\text{rem}2$

$\underline{-8}$

7684 not exactly divisible by 8

2. State whether 277512 is exactly divisible by 8.

Solution

$277\overline{)512}$

$\underline{512} = 64$

$\underline{-8}$

277512, is exactly divisible by 8.

Evaluation Activity



- Without dividing, show whether the following numbers are exactly divisible by 8.
a) 546 b) 2184 c) 7862 d) 8736 e) 43265.
- Without dividing, show whether the following numbers are exactly divisible by 8.
a) 47846 b) 210968 c) 56928 d) 6336 e) 32548.
- Complete this 6-digit number to make it divisible by 8.
7 2 4 □ □ □

Divisibility test of 11

✧ A number is exactly divisible by 11 when the sum of its paired digits from right to left is a multiple of 11 (11, 22, 33, 44, 55, 66, 77, 88, 99, 110, 121, ...)

Examples

1. Without dividing, show whether 36663 is exactly divisible by 11.

Solution

36663

Pairs from right to left

3 [66] [63]

Sum of pairs $3+66+63$

132

132 is a multiple of 11.

Therefore, 36663 is exactly divisible by 11.

2. Without dividing, determine whether 906138 is exactly divisible by 11.

Solution

906138

Pairs from right to left

[90] [61] [38]

Sum of pairs $90+61+38$

189

189 is not a multiple of 11.

Therefore, 906138 is not exactly divisible by 11.

Solution 2

♦ A number is exactly divisible by 11 if the difference between the sum of the digits in the even positions and the sum of the digits in the odd positions is either 0 or a multiple of 11.

1. Without dividing, show whether 36663 is exactly divisible by 11.

Solution

$$3+6+3=12$$

$$3+6+6=15$$

$$6+6=12$$

$$12$$

$$-12$$

$$00$$

Therefore, 36663 is divisible by 11.

2. Without dividing, determine whether 906138 is exactly divisible by 11.

Solution

$$9+6+3=18$$

$$9+0+6+1+3+8=27$$

$$0+1+9=10$$

$$18$$

$$-10$$

$$8$$

Therefore, 906138 is not divisible by 11.

Evaluation Activity



Without dividing, show whether the following numbers are exactly divisible by 11.
a) 62623 b) 154631 c) 13530 d) 42123 e) 81062.

TYPES C

- Wh
- Co
- Oc
- Ev
- Pr
- Co
- Se
- Cu
- Ti

Whole

- ♦ Whol
- 0, 1, 2

Countin

- ♦ They
- ♦ Coun
- 1, 2,

Finding
Con

Exam

1. Pri
wh
So

2. T

TYPES OF NUMBERS

- Whole numbers
- Counting numbers/ natural numbers
- Odd numbers
- Even numbers
- Prime numbers
- Composite numbers
- Square numbers
- Cube numbers
- Triangular numbers



i) Whole numbers

❖ Whole numbers start from zero and keep increasing by 1.

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, ...

ii) Counting numbers/ Natural numbers

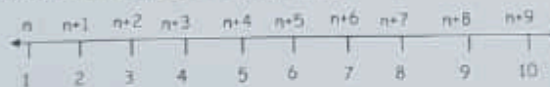
❖ They start from one and keep on increasing by 1.

❖ Counting numbers are consecutive positive integers starting from 1.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, ...

Finding consecutive counting numbers/ natural numbers

Consider positive integers on the number line.



- There is one step interval between counting numbers.
- Two step intervals between the 1st and the 3rd counting numbers.
- Three steps between the first and the fourth counting numbers.

Examples

1. Primary six pupils were asked to write a set of 5 consecutive counting numbers whose median is 8. What numbers did they write?

Solution

1 st	2 nd	3 rd	4 th	5 th
8-2	8-1	8	8+1	8+2

The set is {6, 7, 8, 9, 10}

2. Three rooms were numbered consecutively using counting numbers. If the last room number was 19, find the numbers for the other rooms

Solution

1 st no.	2 nd no.	3 rd no.
19-2	19-1	19

The numbers are 17 and 18

3. The sum of three consecutive counting numbers is 18. Find the numbers.

Solution

Let the first number be k

1 st no.	2 nd no.	3 rd no.	Total
k	$k+1$	$k+2$	18

$$\begin{aligned} k + k + 1 + k + 2 &= 18 \\ k + k + k + 1 + 2 &= 18 \\ 3k + 3 &= 18 \end{aligned}$$

$$\begin{aligned} 3k + 3 - 3 &= 18 - 3 \\ 3k &= 15 \\ \frac{3k}{3} &= \frac{15}{3} \\ k &= 5 \end{aligned}$$

The numbers are:
5, 6 and 7

4. The sum of three consecutive counting numbers is 33. Find the numbers if middle number is y .

Solution

1 st no.	2 nd no.	3 rd no.	Total
$y-1$	y	$y+1$	33

$$\begin{aligned} y - 1 + y + y + 1 &= 33 \\ y + y + y + 1 - 1 &= 33 \\ 3y &= 33 \\ \frac{3y}{3} &= \frac{33}{3} \\ y &= 11 \end{aligned}$$

The numbers are 10, 11 and 12

5. The sum of four consecutive natural numbers is 86. Work out the range if the last number is p .

Solution

1 st	2 nd	3 rd	4 th	Sum
$p-3$	$p-2$	$p-1$	p	86

$$\begin{aligned} p - 3 + p - 2 + p - 1 + p &= 86 \\ p + p + p + p - 3 - 2 - 1 &= 86 \\ 4p - 6 &= 86 \end{aligned}$$

$$4p - 6 + 6 = 86 + 6$$

$$\begin{aligned} 4p &= 92 \\ \frac{4p}{4} &= \frac{92}{4} \\ p &= 23 \end{aligned}$$

The numbers are 20, 21, 22 and 23

Range
23 - 20
= 3

Evaluation Activity



- Find the three consecutive counting numbers whose median is 14.
- Find the sum of five consecutive natural numbers whose median is 15.
- The total of three consecutive positive integers is 45.
 - Find the numbers.
 - Calculate their average.
- The sum of three consecutive counting numbers is 138. If the middle number is
 - Find the numbers.
 - Calculate their range.

5. The sum of four consecutive counting numbers is 54. If the last number is p , find the numbers.
6. In three days, a farmer planted 240 trees. He planted one more tree each day than the previous day. How many trees did he plant each day?
7. Sharon sold 48 litres of milk in 3 hours. She sold one more litre of milk each hour than the previous hour.
 - a) Find the number of litres she sold in the first hour.
 - b) How many more litres of milk did she sell on the last day than the first day?

iii) Odd numbers

- ❖ These are numbers that give a remainder when divided by 2.
 - ❖ An odd number is a counting number which is exactly not divisible by 2.
 - ❖ They start from one and keep on increasing by 2.
- $\{1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, \dots\}$

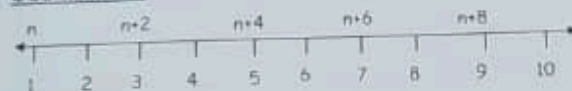
iv) Even numbers

- ❖ These are numbers that are exactly divisible by two.
 - ❖ Even numbers are whole numbers which are exactly divisible by 2.
 - ❖ They start from zero and keep increasing by 2.
- $\{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, \dots\}$

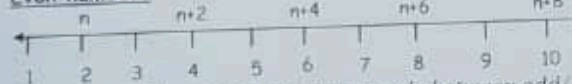
Finding consecutive odd and even numbers

Consider the number lines below for odd and even numbers

Odd numbers



Even numbers



- ✓ There are two step intervals between odd/even numbers.
- ✓ Two step intervals between the 1st and the 3rd number on both lines.
- ✓ Six steps between the first and the fourth number on both lines.

Examples

1. The median of five consecutive odd numbers is 7, find the numbers.

Solution

1 st	2 nd	3 rd	4 th	5 th
$7-4$	$7-2$	7	$7+2$	$7+4$

The numbers are 3, 5, 7, 9, 11

2. Three cards were numbered consecutively using even numbers. If the last card number was 48, find the numbers for the other cards.

Solution

1 st no.	2 nd no.	3 rd no.
$48-4$	$48-2$	48

The numbers are 44 and 46



3. The sum of three consecutive even numbers is 18. Find the numbers.

Solution:

Let the first number be k

1 st no.	2 nd no.	3 rd no.	Total
k	$k+2$	$k+4$	18

$$k + k + 2 + k + 4 = 18$$

$$3k + 6 = 18$$

$$3k + 6 - 6 = 18 - 6$$

$$3k = 12$$

$$\frac{3k}{3} = \frac{12}{3}$$

$$k = 4$$

The numbers are 4, 6 and 8

4. The sum of three consecutive odd numbers is 33. Find the numbers if the first number is n .

Solution:

1 st no.	2 nd no.	3 rd no.	Total
$n-2$	n	$n+2$	33

$$n - 2 + n + n + 2 = 33$$

$$3n = 33$$

$$\frac{3n}{3} = \frac{33}{3}$$

$$n = 11$$

The numbers are 9, 11 and 13

5. The sum of four even numbers is 84. Work out the range of the numbers if the last number is p .

Solution:

1 st	2 nd	3 rd	4 th	Sum
$p-6$	$p-4$	$p-2$	p	84

$$p - 6 + p - 4 + p - 2 + p = 84$$

$$4p - 12 = 84$$

$$4p - 12 + 12 = 84 + 12$$

$$4p = 96$$

$$\frac{4p}{4} = \frac{96}{4}$$

$$p = 24$$

The numbers are 18, 20, 22 and 24

Range

$$24 - 18$$

$$= 6$$

Evaluation Activity



- Find the three consecutive odd numbers whose median is 19.
- Find the sum of five consecutive even numbers whose median is 16.
- The total of three consecutive odd is 45.
 - Find the numbers.
 - Calculate their average.
- The sum of three consecutive even numbers is 78. If the middle number is n .
 - Find the numbers.
 - Calculate their range.
- The sum of four consecutive odd numbers is 88. If the last number is p find the numbers.

6. The sum of 3 consecutive even numbers is 126. If the first number is $(4n+20)$, find the number.
7. Achai types 150 pages of a book in 3 hours. If she types two more pages each hour than the previous one, how many pages did she type in the last hour?
8. Okello scores 10 more marks in each exam than the previous one. If he scored 390 marks in 6 sets of exams, how many marks did he score in the fourth set?

v) **Prime numbers**

- ❖ These are counting numbers with only two factors.
- ❖ The factors are one and the number itself

Examples

Prime number	2	3	5	7	11	13	17	19	23	29
Factors	(1,2)	(1,3)	(1,5)	(1,7)	(1,11)	(1,13)	(1,17)	(1,19)	(1,23)	(1,29)

vi) **Composite numbers**

- ❖ These are numbers with more than 2 factors.

Examples:

Composite number	Factors	Composite number	Factors
4	(1,2,4)	12	(1,2,3,4,6,12)
6	(1,2,3,6)	14	(1,2,7,14)
8	(1,2,4,8)	15	(1,3,5,15)
9	(1,3,9)	16	(1,2,4,8,16)
10	(1,2,5,10)	18	(1,2,3,6,9,18)

Note:

- ✓ All non-prime numbers apart from 0 and 1 are composite numbers.

vii) **Square numbers**

- ❖ These are numbers got by multiplying a counting number by its self.

Examples:

Square number	1	4	9	16	25	36	49	64	81	100	121
Multiplication	1×1	2×2	3×3	4×4	5×5	6×6	7×7	8×8	9×9	10×10	11×11

- ❖ Square numbers can also be got by adding consecutive odd numbers.

Examples

Addition	Square number	Addition	Square number
1	1	$1+3+5+7+9+11$	36
$1+3$	4	$1+3+5+7+9+11+13$	49
$1+3+5$	9	$1+3+5+7+9+11+13+15$	64
$1+3+5+7$	16	$1+3+5+7+9+11+13+15+17$	81
$1+3+5+7+9$	25	$1+3+5+7+9+11+13+15+17+19$	100

viii) Cube numbers

These are got by multiplying the same number thrice

Examples:

Multiplication	cube number	multiplication	cube number
$1 \times 1 \times 1$	1	$6 \times 6 \times 6$	216
$2 \times 2 \times 2$	8	$7 \times 7 \times 7$	343
$3 \times 3 \times 3$	27	$8 \times 8 \times 8$	512
$4 \times 4 \times 4$	64	$9 \times 9 \times 9$	729
$5 \times 5 \times 5$	125	$10 \times 10 \times 10$	1000

ix) Triangular numbers

These are numbers got by adding consecutive counting numbers.

Examples:

Addition	Triangular number	Addition	Triangular number
1	1	$1+2+3+4+5+6$	21
$1+2$	3	$1+2+3+4+5+6+7$	28
$1+2+3$	6	$1+2+3+4+5+6+7+8$	36
$1+2+3+4$	10	$1+2+3+4+5+6+7+8+9$	45
$1+2+3+4+5$	15	$1+2+3+4+5+6+7+8+9+10$	55

Triangular numbers can also be found using the formula:

$$\frac{n(n+1)}{2}$$

Where n is the position of the required triangular number.

Example

Find the ninth triangular number.

Solution:

$\frac{n(n+1)}{2}$	$\frac{9 \times 10}{2}$	45
$\frac{9(9+1)}{2}$	$\frac{9 \times 10}{2}$	

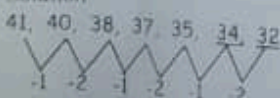
Completing sequences

Examples

Find the next two numbers in the sequence below.

41, 40, 38, 37, 35, __, __

Solution



Mixed Activity



1. Find the next number in the series.
 - a) 2, 3, 5, 7, ____
 - b) 1, 3, 5, 7, ____
 - c) 4, 6, 8, 9, 10, ____
 - d) 49, 36, 25, 16, ____
 - e) 625, 125, 25, ____
2. What is the product of the 3rd and the 6th triangular numbers?
3. Write the seventh prime number in Roman numerals
4. What is the difference between the 10th odd number and the 10th even number?
5. Work out the total of the first 8 triangular numbers.
6. Find the sum of the 8th and the 11th triangular number.
7. Workout the difference between the 6th composite number and the 4th square number.

